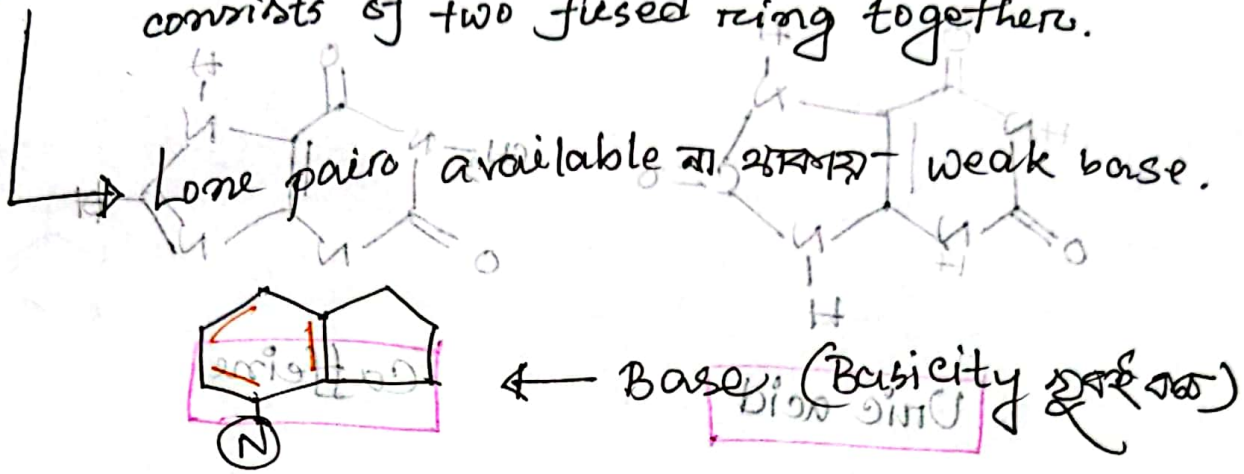
Purine & Nucleic Acid

Date: 23.05.24
Suzon-50

F-22, 18, 4

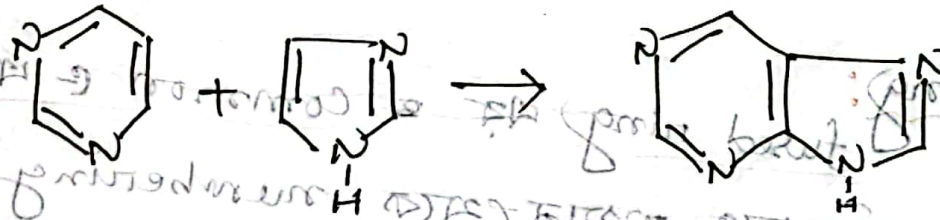
Purine: Organic heterocyclic aromatic compound that consists of two fused ring together.



→ Lone pair available বা সহজলভ্য → weak base.

← Base (Basicity হ্রাস পায়)

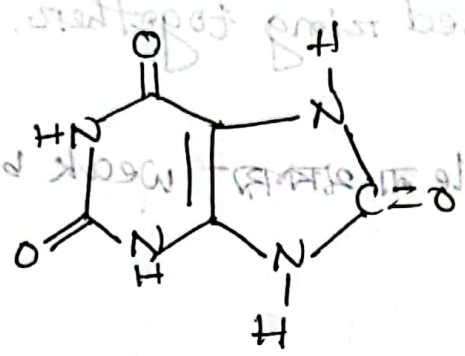
Purin is the parent structure of a group of cyclic diuriedes and was dis covered by Emil Fischer. It is a hetero aromatic compound consist of a pyrimidine ring fused with an imidazole ring.



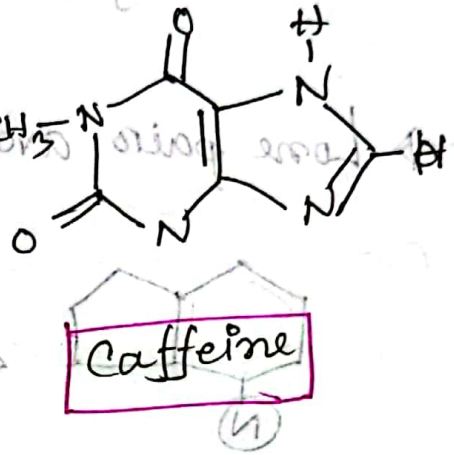
Pyrimidine + Imidazole → Purin

Nuclcic Acid : DNA, RNA

Derivatives of Purines:

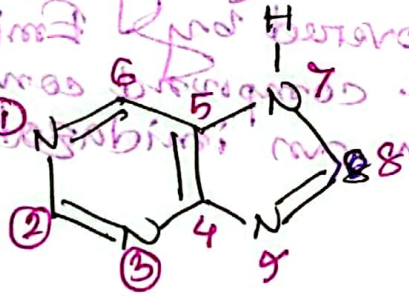


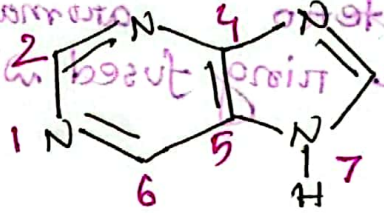
Uric acid



Caffeine

Two tautomeric form of purine

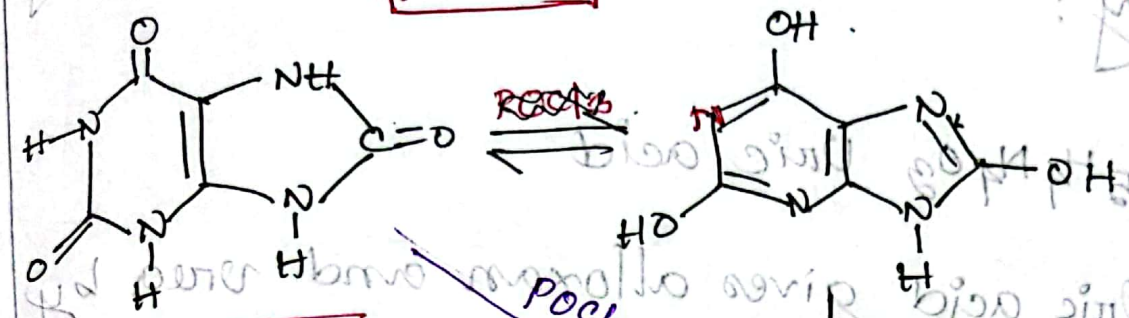




Numbering: fused ring এর ক্ষেত্রে common ও এরূপ যে পাশে N-গুলো দূরে সেখান থেকে numbering শুরু করা যেদিকে N আছে সেইদিকে যেতে হবে। Then, পরের ring-র যে N এর পাশে H যুক্ত সেখান থেকে শুরু করতে পর হবে।

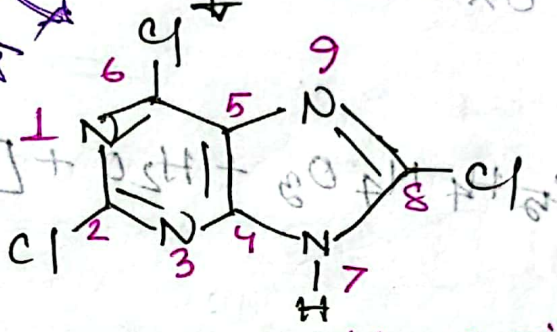
Synthesis of Purine:

E-18

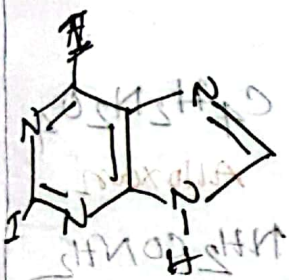


Uric acid

$POCl_3$, 4 hr



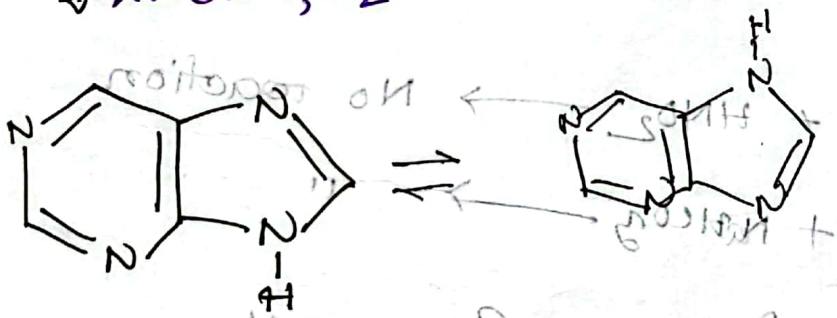
2,6,8 – trichloropurin.



HI, PH_4I

6°C

100°C, Zn dust, H_2O



Purine

Polymer of Nucleotides

Nucleoside + PO_4^{3-}

Base + Sugar (Deoxyribose/ribose)

(Purin/Pyrimidine)

#Find out the structure of Uric acid by degradation study:

A Final Q

1. $C_5H_4N_4O_3$ Uric acid

2. Uric acid gives alloxan and urea by oxidation with HNO_3.

$$C_5H_4N_4O_3 + H_2O + [O] \xrightarrow{HNO_3} C_4H_2N_2O_4 + NH_2CONH_2$$

Alloxan + Urea

$$C_4H_2N_2O_4 + HNO_2 \longrightarrow \text{No reaction}$$

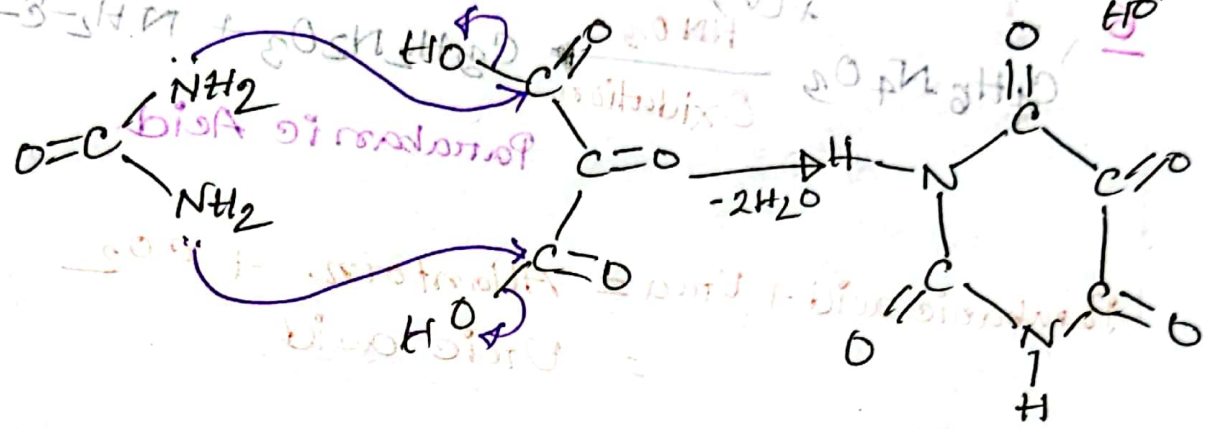
$$\text{"} + NaHCO_3 \longrightarrow \text{"}$$

Therefore, No free $-NH_2$ & $-COOH$

(3) In alloxan, there is no free amino or carboxylic acid group when alloxan is hydrolysed with alkali it produces one mole of Urea and one molecule of mesoxalic

Oxalic acid.

$$C_4H_2N_2O_4 + H_2O \xrightarrow{KOH} NH_2-C(=O)-NH_2 + \text{mesoxalic acid}$$



Addition of one mole of Urea with alloxane gives the structure of uric acid.

Q-1: 1 mole Urea + 1 mole meso oxalic acid = Alloxane – Explain.

Synthesis of important derivative of Purin Base:

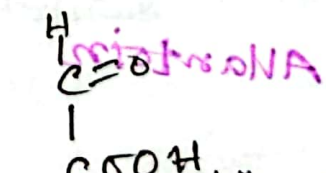
④ $C_5H_4N_4O_3 \xrightarrow[\text{Hydrolysis}]{+H_2O+[O]\ KMnO_4} C_4H_6N_4O_3 + CO_2$

Allantoin.

F-20

⑤ $C_4H_6N_4O_3 \xrightarrow[\text{Hydrolysis}]{KOH} 2NH_2-C(=O)-NH_2 + \text{glyoxalic acid}$

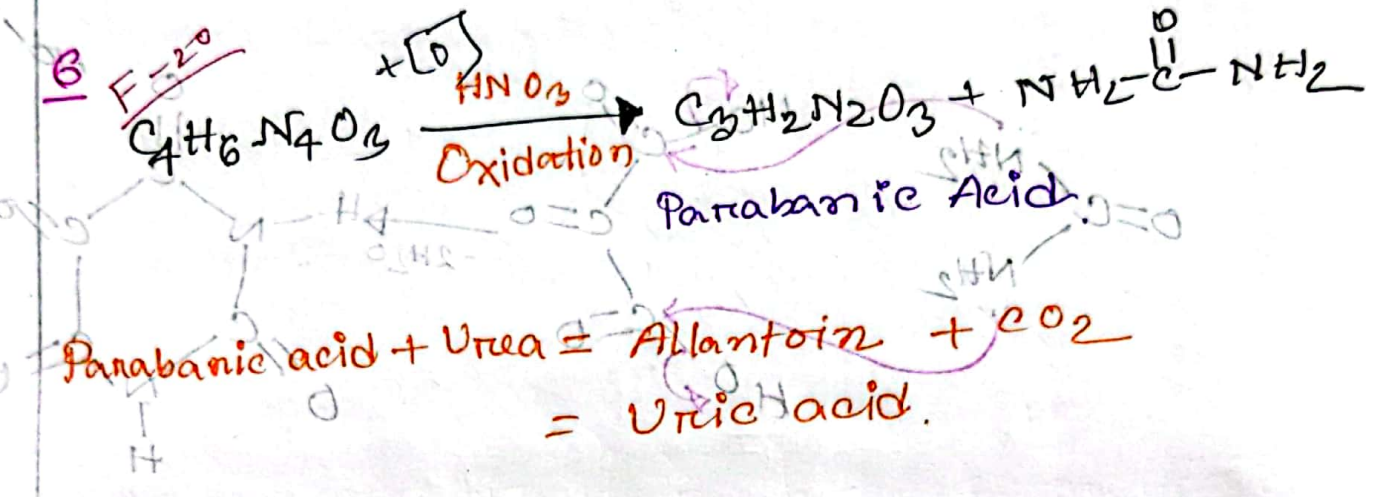
$+2H_2O$



Glyoxalic acid.

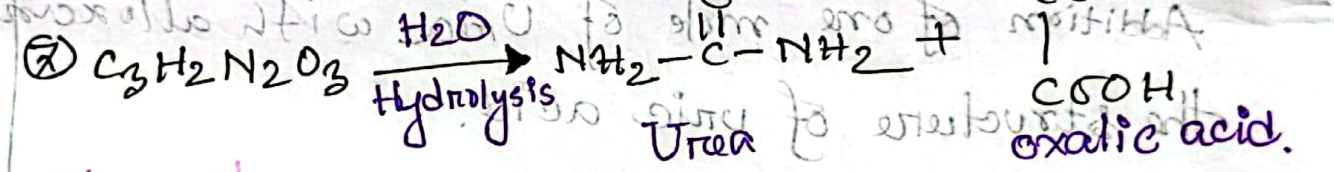
Q: Allantoin is Diureide of Glyoxalic acid – explain

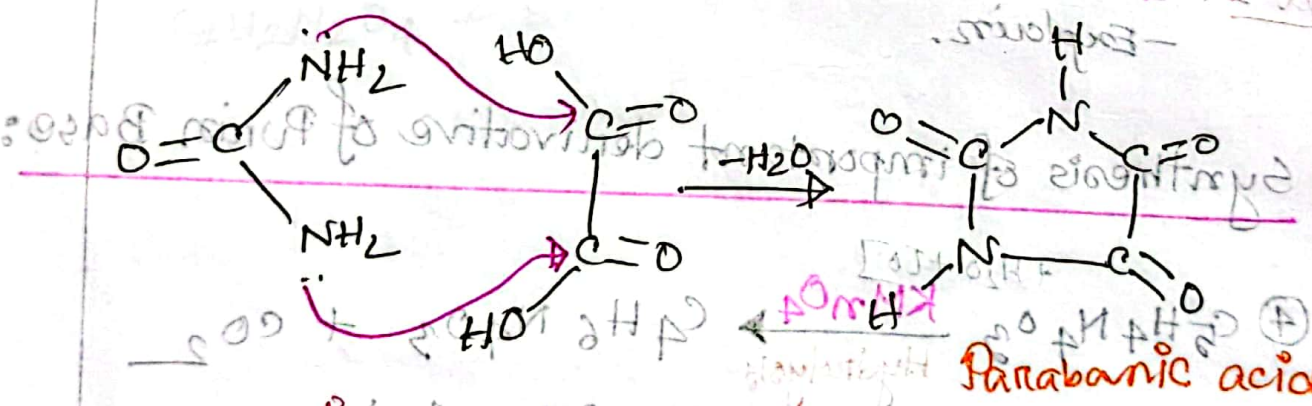
Soln: Above step.

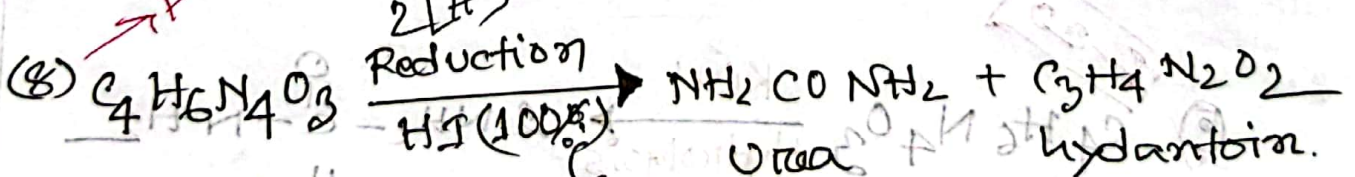


Parabanic acid + Urea = Allantoin + CO_2

= Uric acid.



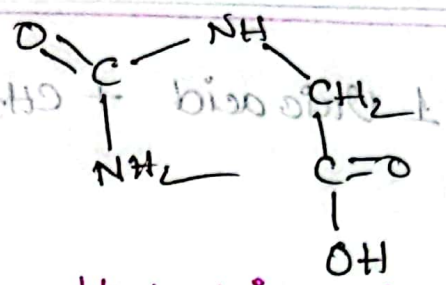


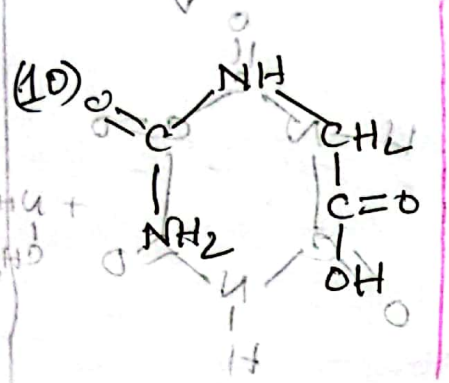


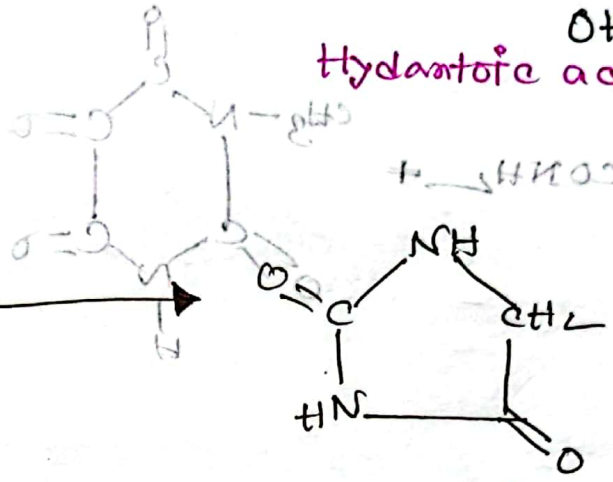
Allantoin

SUZON-50

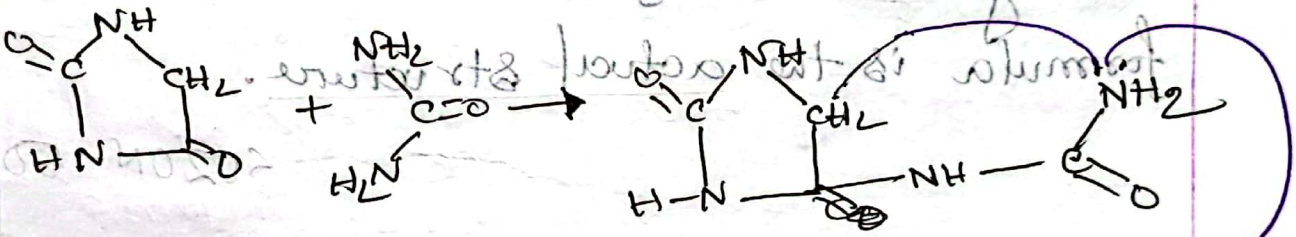
⑨ $C_5H_4N_2O_3$ ⟵ (Hydrolysis, H_2O) ⟶ Hydantoic acid.

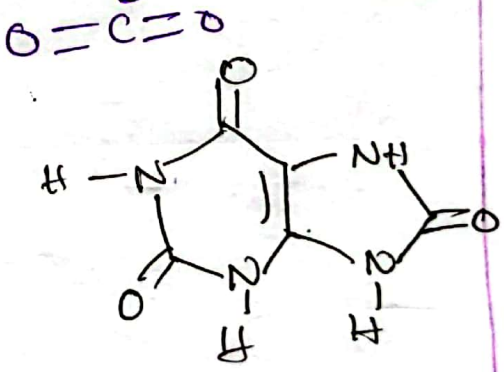


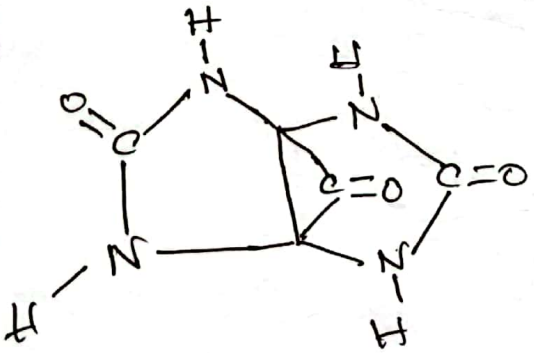




Hydantoin







Fitting formula

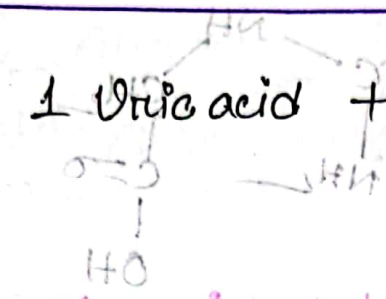
Actual structure

Medicus formula

Medicus

F-21 How can you prove that medicus structure of uric acid is the actual structure? (F-21)

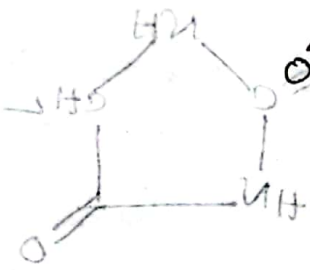
1 Uric acid + CH_3I $\xrightarrow{NaOH}$ A + B

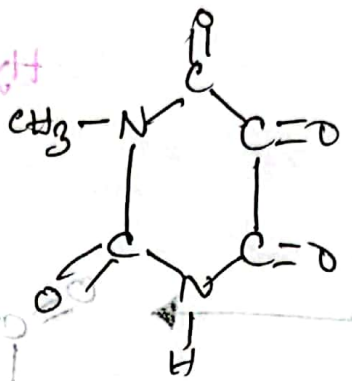


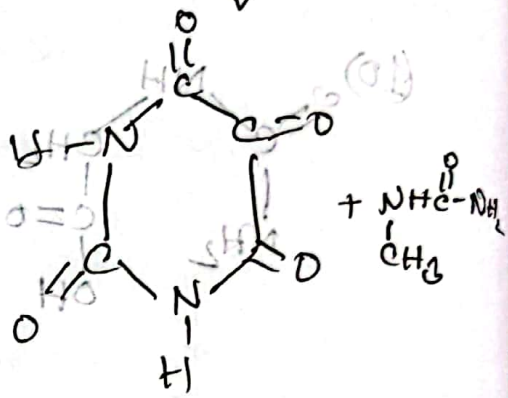
HNO_3

HNO_3

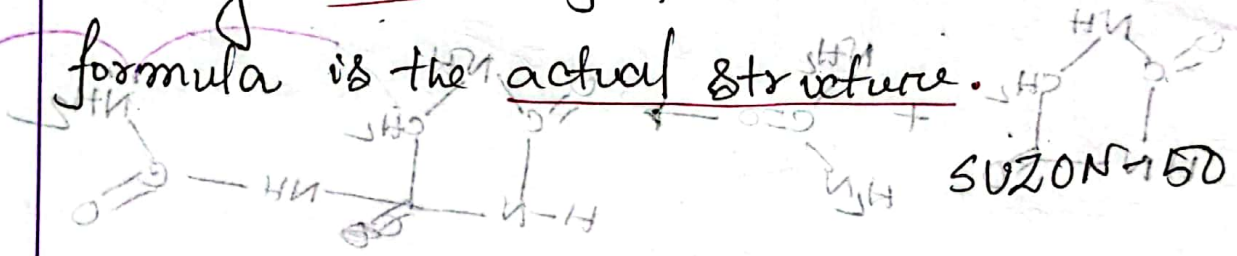
NH_2CONH_2 +





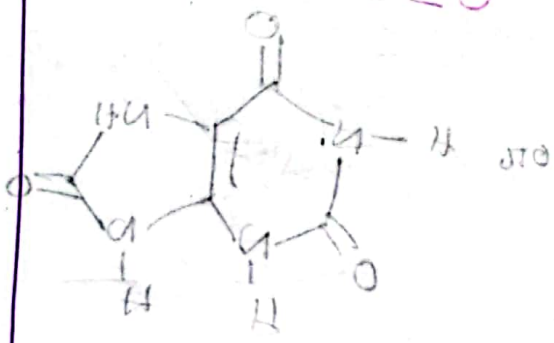


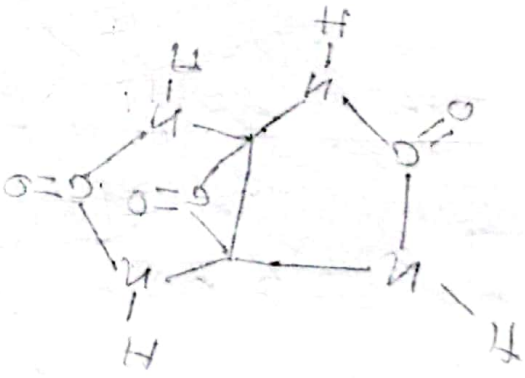
We get mixture of product. So medicus formula is the actual structure.



SUZON-150







Medicus formula

Actual structure

Fitting formula

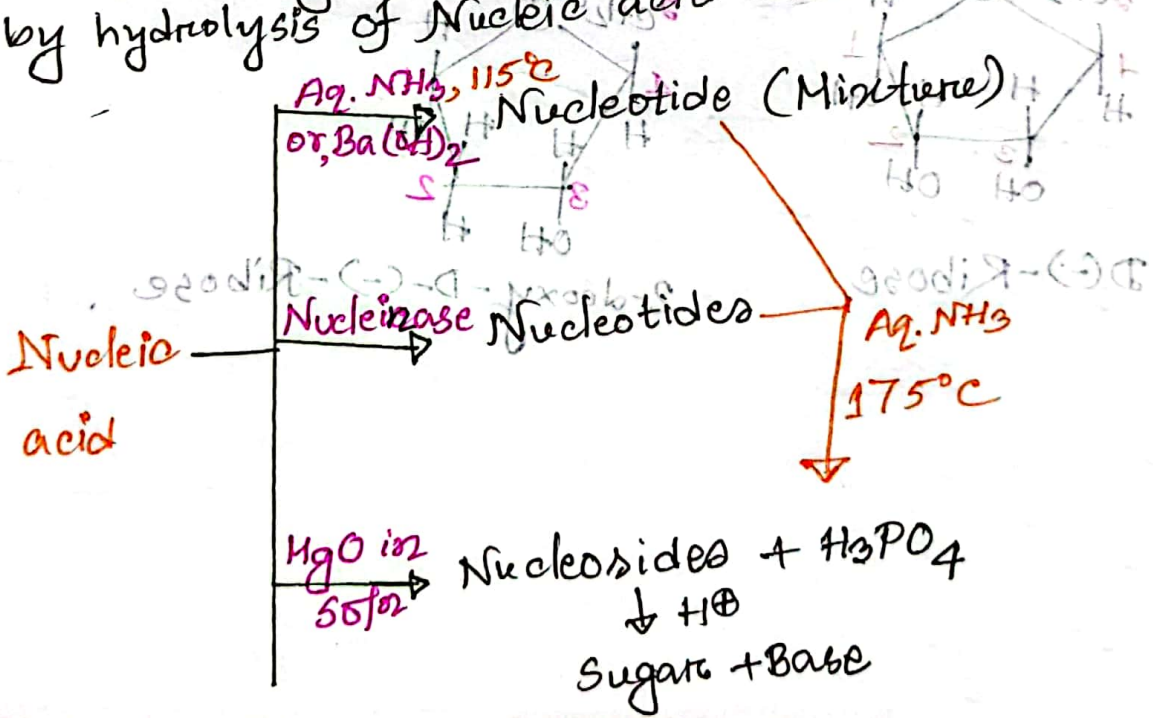
Nuclei acid : Nucleic acids are high molecular weight polymer, colourless amorphous solid. They are polymers of nucleotides.

Nucleic acid is consist of base (Purine and pyrimidine) and pentose sugar (Ribose + 2-deoxyribose) attached with an inorganic H_3PO_4 group.

Nucleic acid (Polymer of Nucleotides)
- B + S + P → Nucleotide.
- B + S + P
- B + S + P
- B + S → Nucleoside.

F-21

Q. The following chart shows the nature of products obtained by hydrolysis of Nucleic acid under different conditions



Nucleic acid ⟶ Polymer of Nucleotide

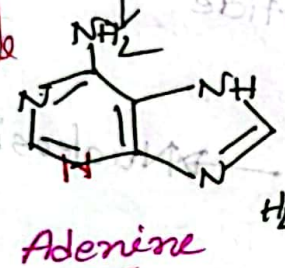
Nucleotides ⟶ Nucleosides + H_3PO_4.

Nucleoside ⟶ Pentose sugar + Base (Ribose/2-Deoxy

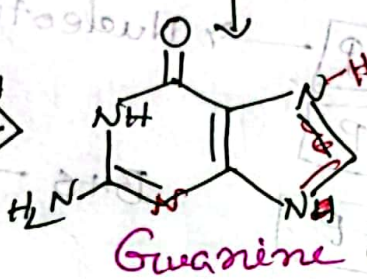
+

Purin + Pyrimidine.

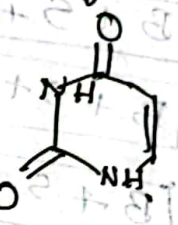
F-20 Name + structure



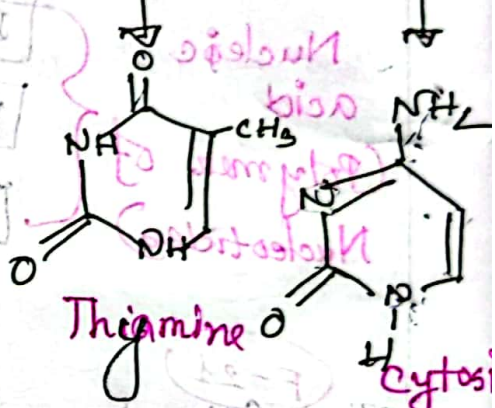
Adenine



Guanine

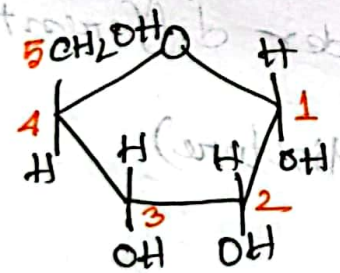


Uracil

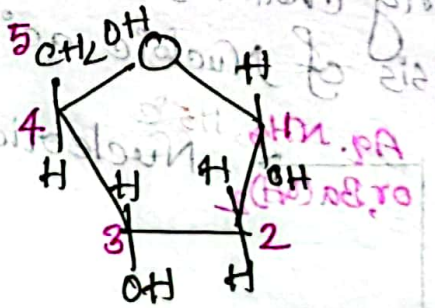


Thiamine

Cytosi



D(-)-Ribose

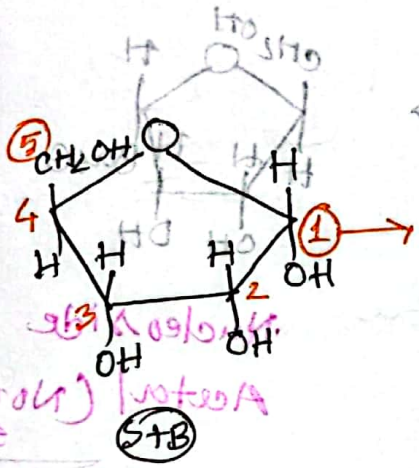


2-deoxy-D-(-)-Ribose

Adenosine (Ribose + adenine)

Guanosine (Guanine + Ribose)

Thyamidine (Thyamine + 2-deoxyribose)

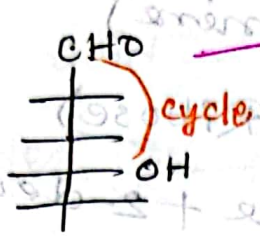


Anomaric Carbon

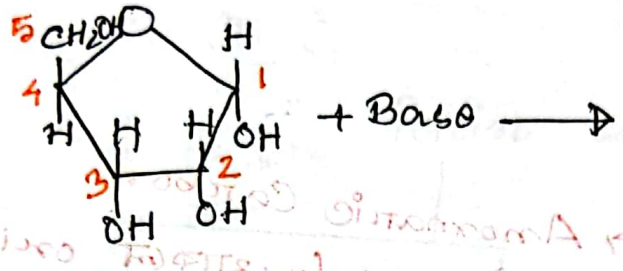
C-এর OH free থাকলে oxidizing agent দ্বারা oxidized হয় reducing sugar বলে। যদি OH free না থাকে তখন Non-reducing sugar বলে।

SUZON

CHEM-50

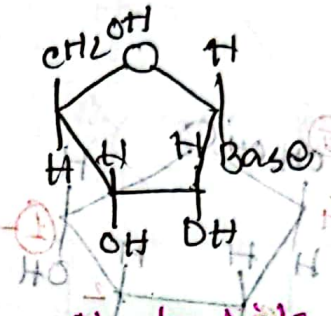


→ Reducing sugar.



Ribose sugar

Hemiacetal (Reducing sugar)



Nucleoside

Acetal (Non-reducing sugar)

Why?

Fehling & Tollens reagent

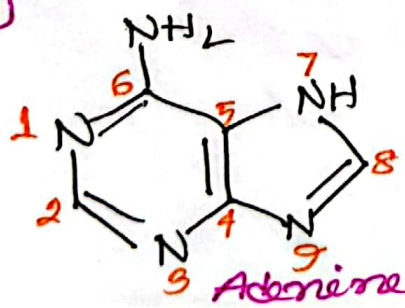
Mild oxidizing agent.

Q→ Sugar এর কয় নম্বরে Base যুক্ত আছে? কীভাবে explain করবে?

উঃ: 1-No C. Explanation→above.

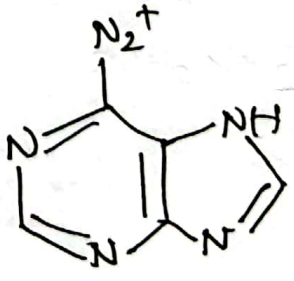
Q→ In which position of Base the sugar is attached?

①

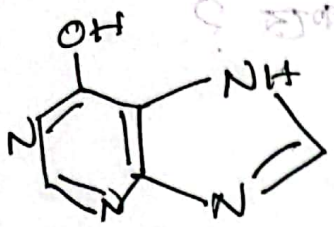


Adenine.

+ HNO_2 →

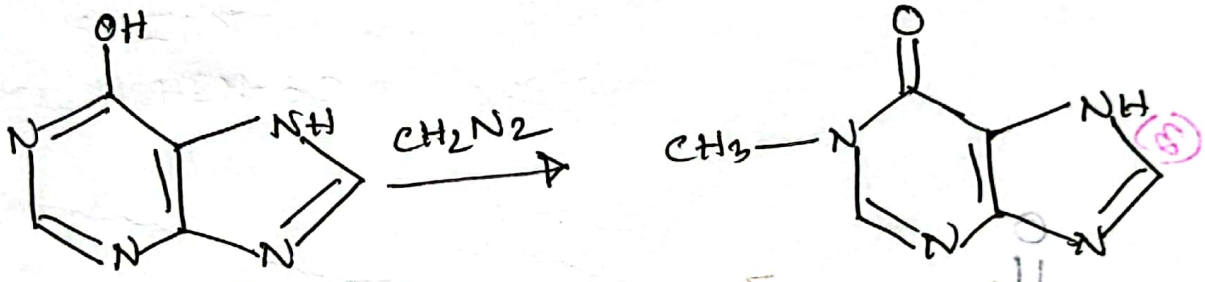


→ H_2O



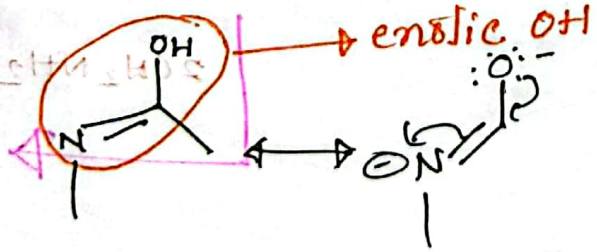
When adenine is treated with HNO_2 the NH_2(6) group is 6-C is deaminated and evolve N_2 gas at room temperature in presence of H_2O and replace NH_2(6) by -OH group. So, 1° NH_2 is not attached to the sugar (C-1).

(2)



When phenol derivative is treated with diazomethane (CH_2N_2) the $-CH_3$ group attack at N-1

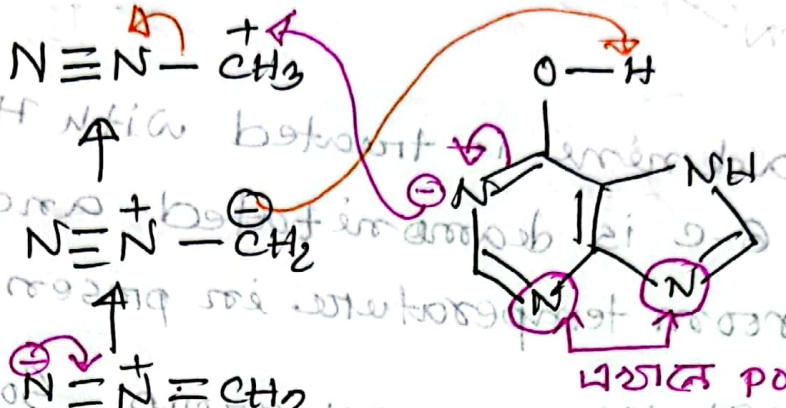
No sugar at N-1 of base.



**

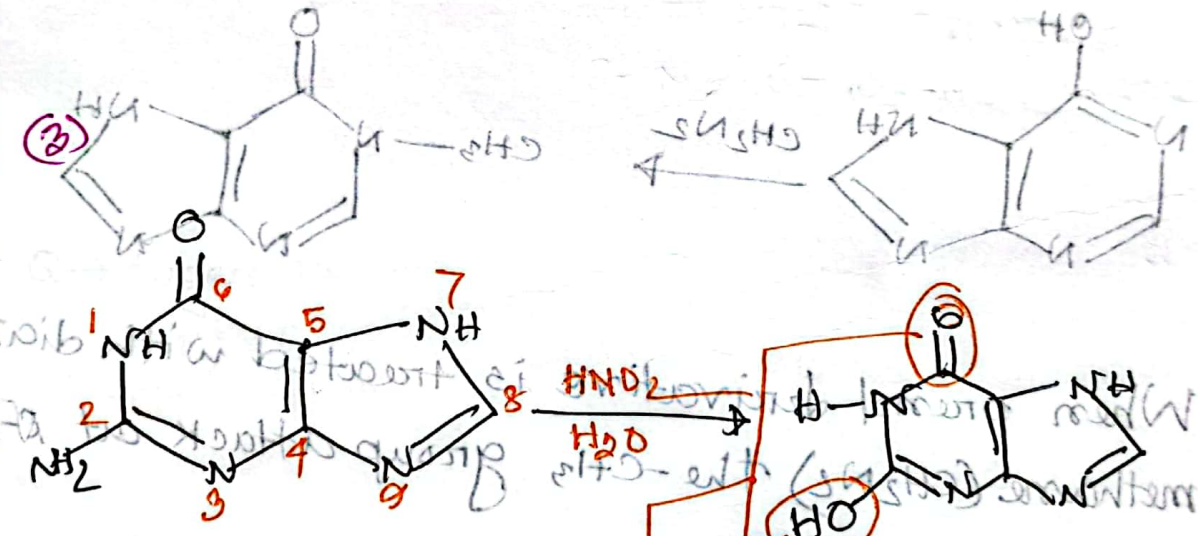
Q: Diazomethane কোন N attack করে ?

Solⁿ:

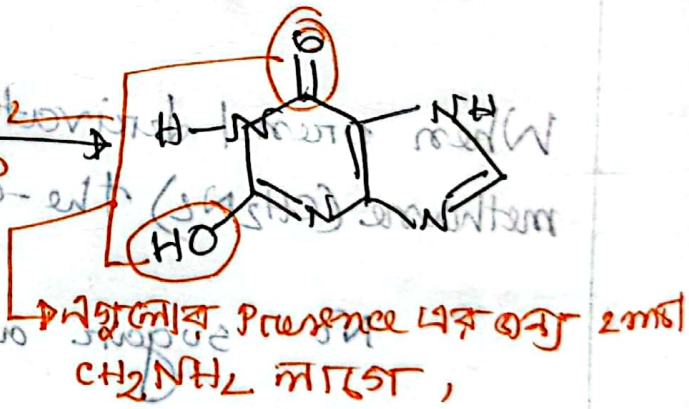


Mechanism

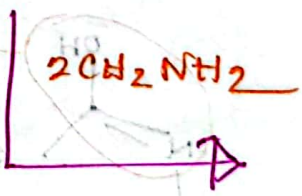
Therefore, 1- N attack করে ।

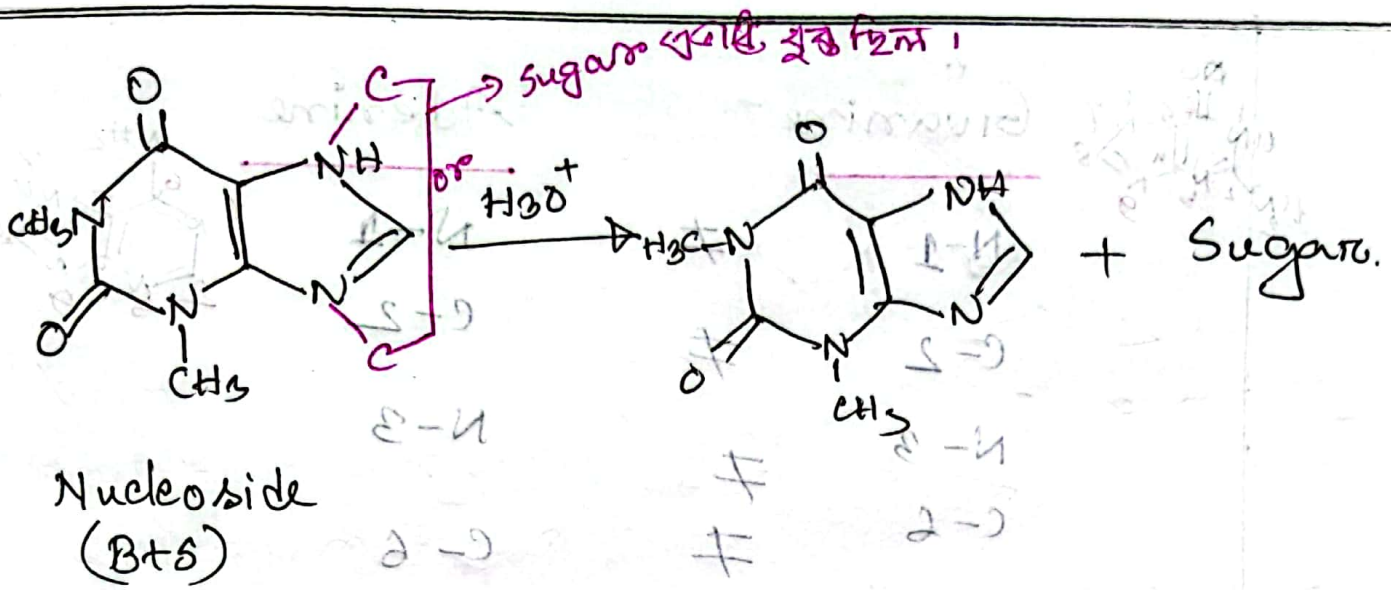


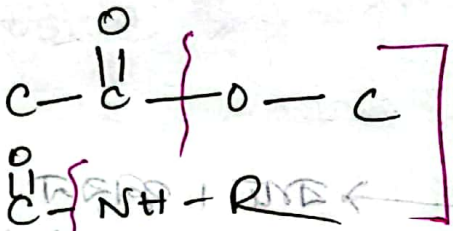
Guanosine
↓
Nucleoside (S+B)



এখানে possible N, দুটোর Presence এর জন্য 2 মোল CH_2NH_2 লাগে,

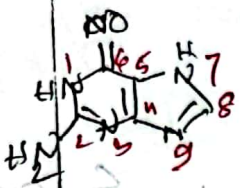




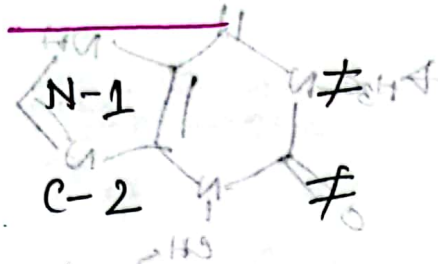


Though the two methyl groups are attached at C-1 and C-3 positions. So, the sugar is not attached at C-1 or C-3 position.

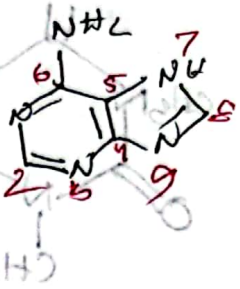
④ Enzyme reacts in a specific site. The sugar is attached in the same position of adenine and Guanine. The equivalent and nonequivalent of adenine and Guanine are —



Guanine



Adenine



Guanine		Adenine
N-1	≠	N-1
C-2	≠	C-2
N-3	≠	N-3
C-6	≠	C-6

N-7 ≈ N-7

C-8 ≈ C-8 → বাকি ৪ কারন C-C bond কে Hydrolysis এর মাধ্যমে ভাঙা যায় না,

N-9 ≈ N-9

C-N / C-O → ভাঙাভাঙি হবে,

(4)

Nucleotide → Nucleoside + H_3PO_4

↓

Base + Sugar (C-1) → Ribose

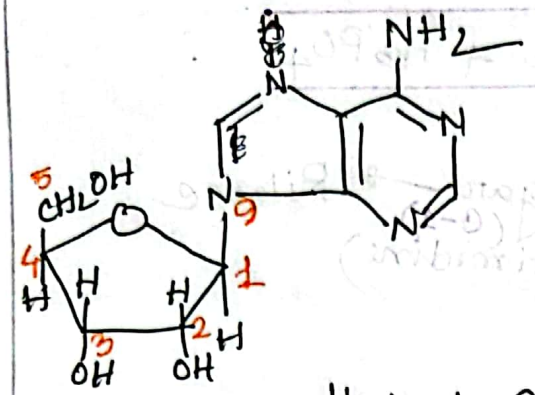
(Purin/pyrimidine)

If the UV spectroscopy applied the λ max for nucleoside Guanosine (Guanine + Ribose) is found 463 nm and if the N-9 H is replaced by $-CH_3$ then the λmax is 460 nm which is almost same as the guanosine.

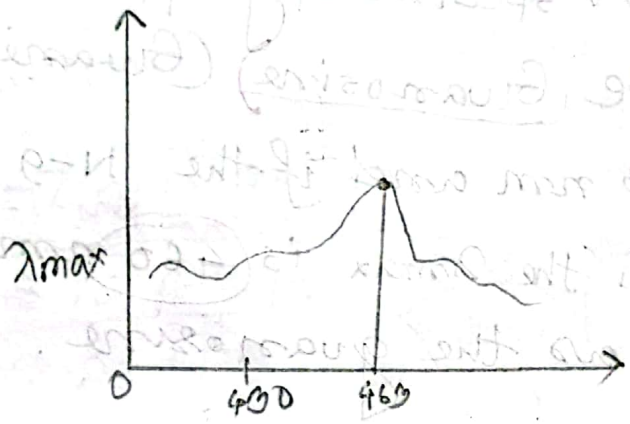
But if the N-7, H is replaced by $-CH_3$ we get,

λmax at 440 nm which is different from the guanosine.

So, the sugar (C-1) is attached at N-9 of purin base.



Highest λmax 463 nm absorb করে।



Chromophorosity → Conjugate double bond অথবা system বলিয়ে দেয়।

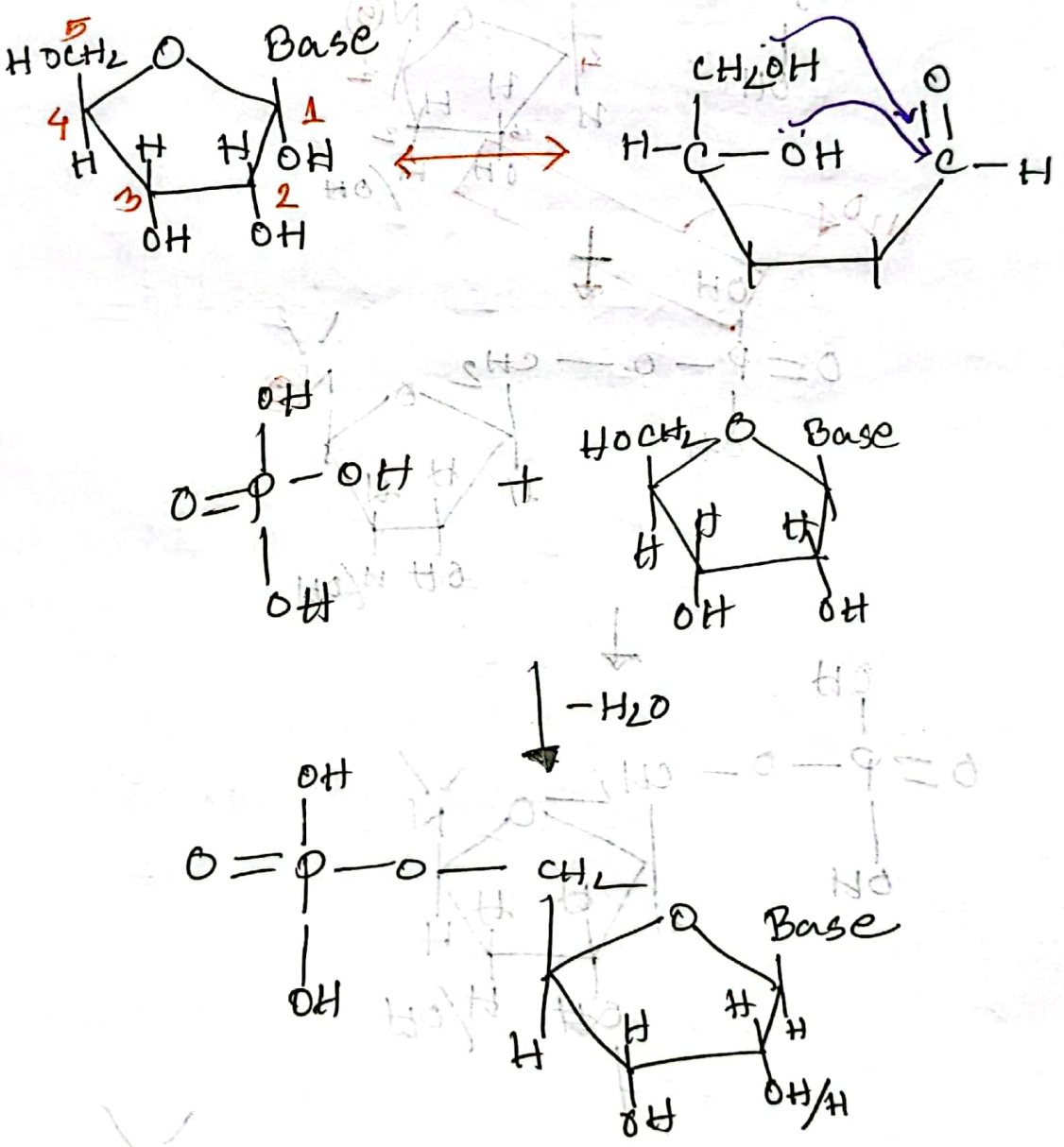
In which part of Nucleoside the PO_4 is attached and How?

When nucleotide is hydrolysed under controll environment we get ribose monophosphate and base.

Nucleotides $\xrightarrow[\text{Hydrolysis}]{\text{Control}}$ Ribose monophosphate + Base

↳ এর মানে PO_4 এর সাথে যুক্ত নয়।

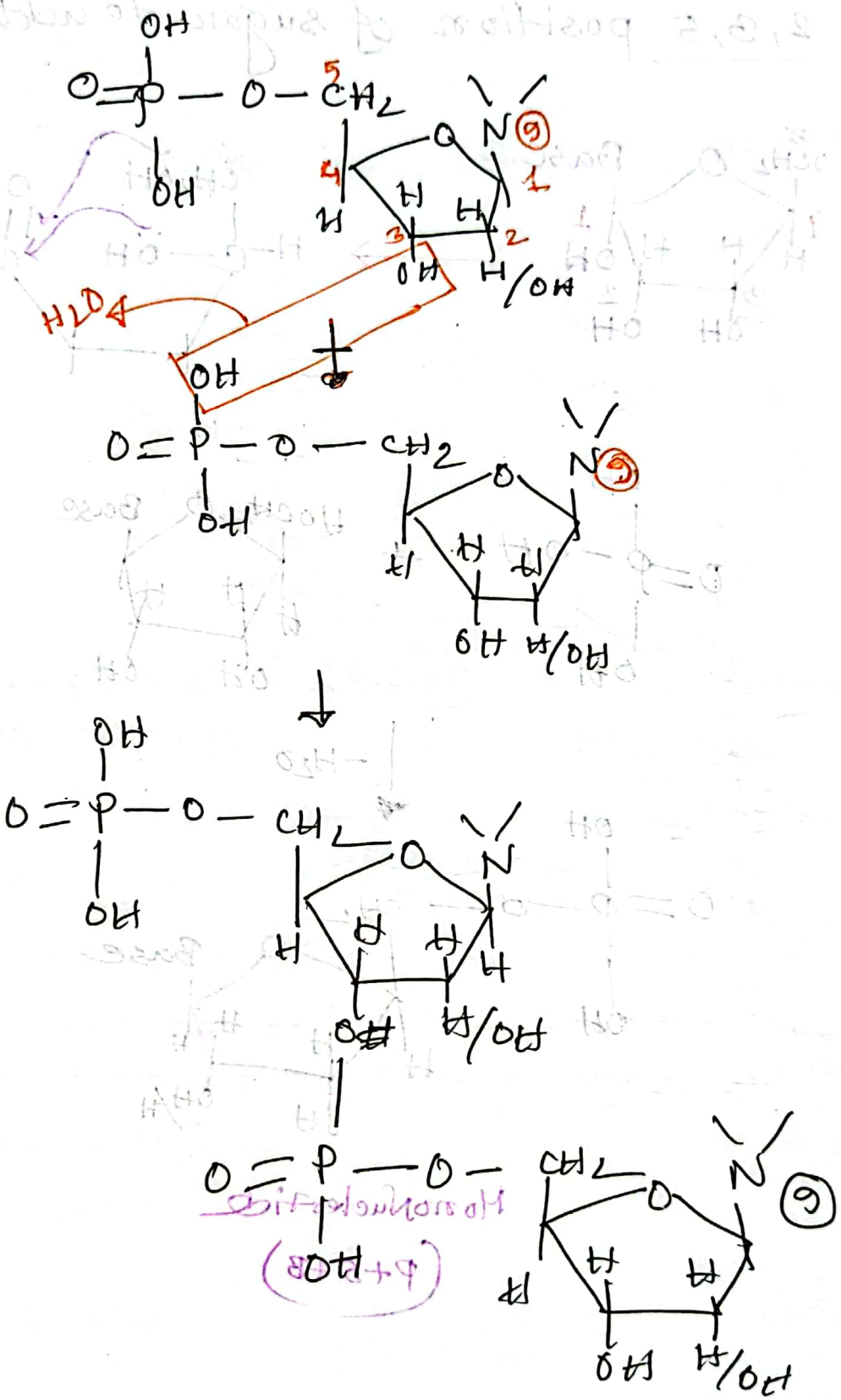
So the $-PO_4$ group is attached with the sugar part. The sugar contains $-OH$ group at 2, 3, 5 positions. So the addition of H_3PO_4 will attack at 2,3,5 position of sugar to add (PO_4) group.



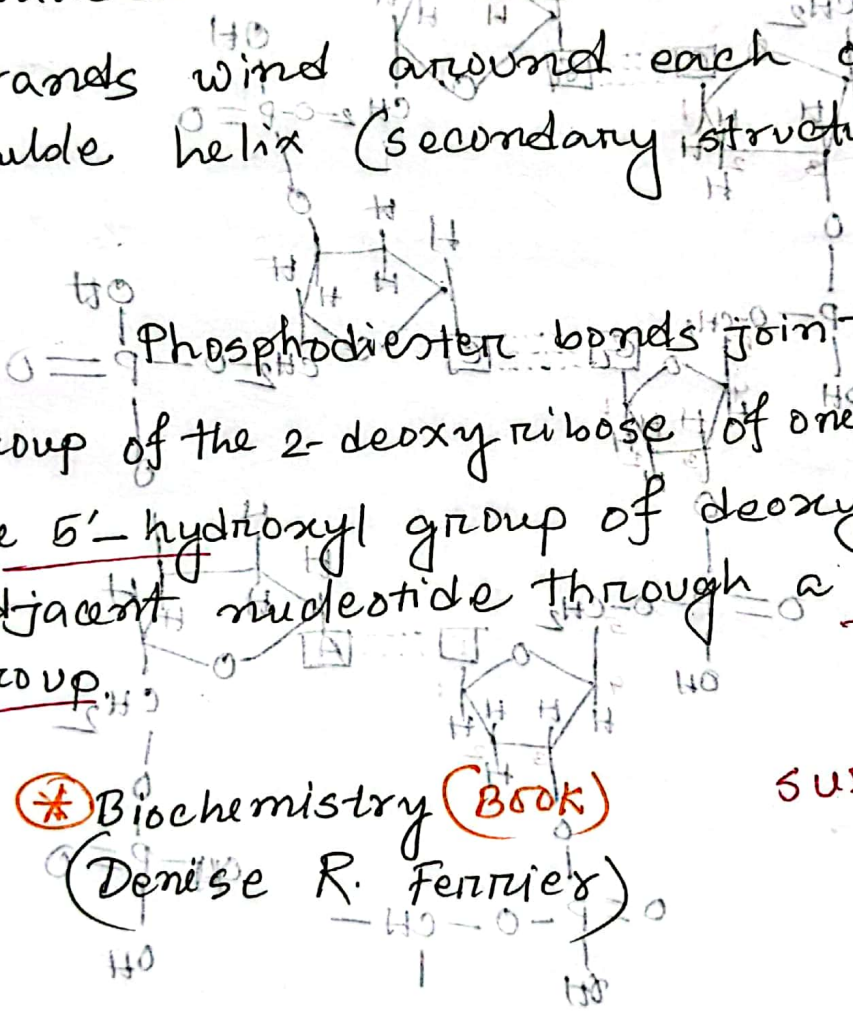
Mononucleotide

(P+S+B)

পরপর অনেকগুলো Mononucleotide যুক্ত হয়ে Polynucleotide হয়।



*DNA is a polymers of deoxyribonucleic monophosphoric ate (dNMP) covalently linked by 3'→5' phosphodiester bonds. DNA exist as double stranded molecule (dsDNA) in which two strands wind around each other forming double helix (secondary structures)

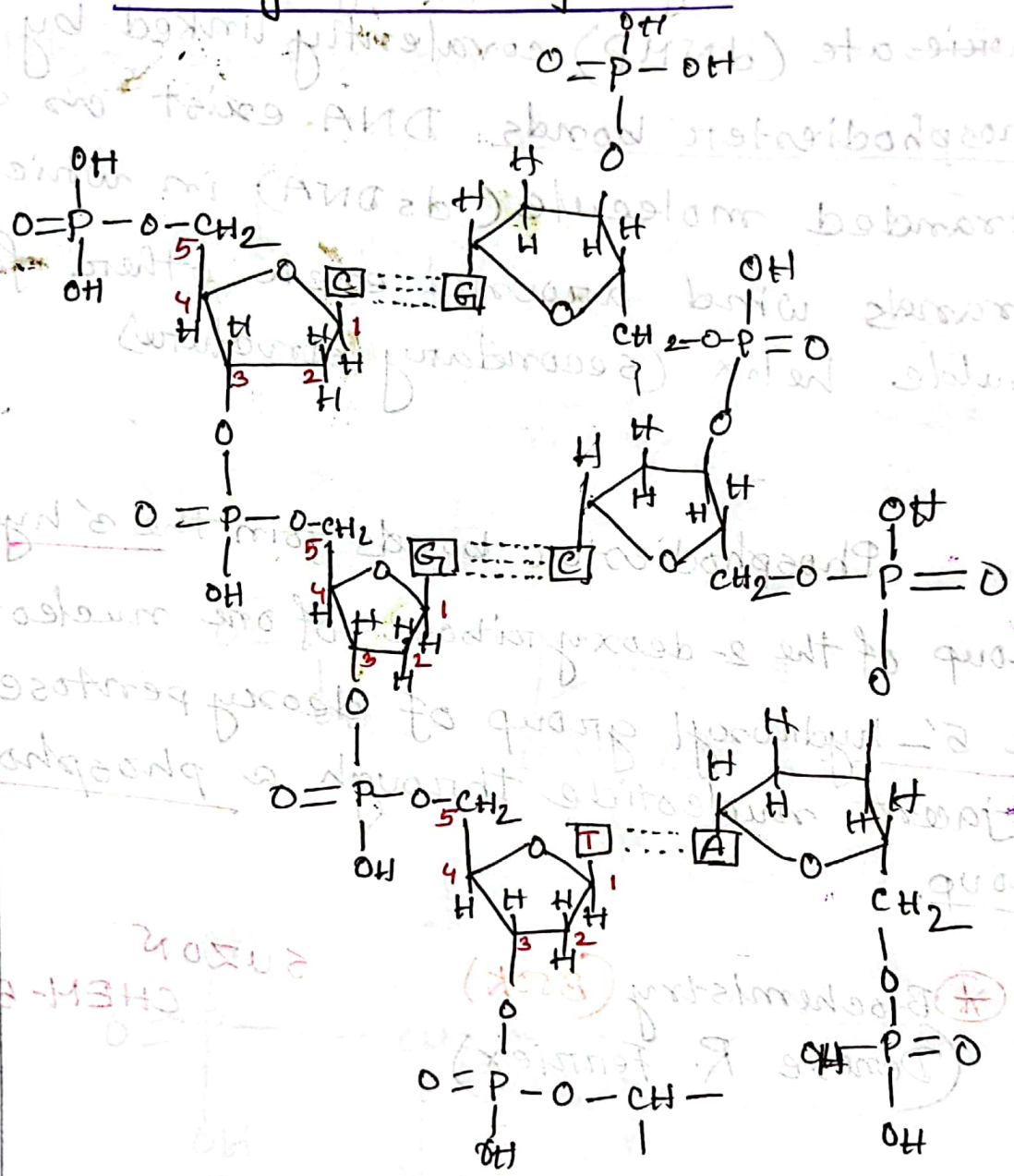


Phosphodiester bonds join the 3' hydroxyl group of the 2-deoxyribose of one nucleotide to the 5'-hydroxyl group of deoxypentose of an adjacent nucleotide through a phosphoryl group.

*Biochemistry (BOOK)
(Denise R. Ferrier)

SUZON
CHEM-50

Primary structure of DNA



H-bond হলে তখন 20Å distance এ হয়। Purin-Purin distance বেশি হয় তাই H-bonding হয় না।

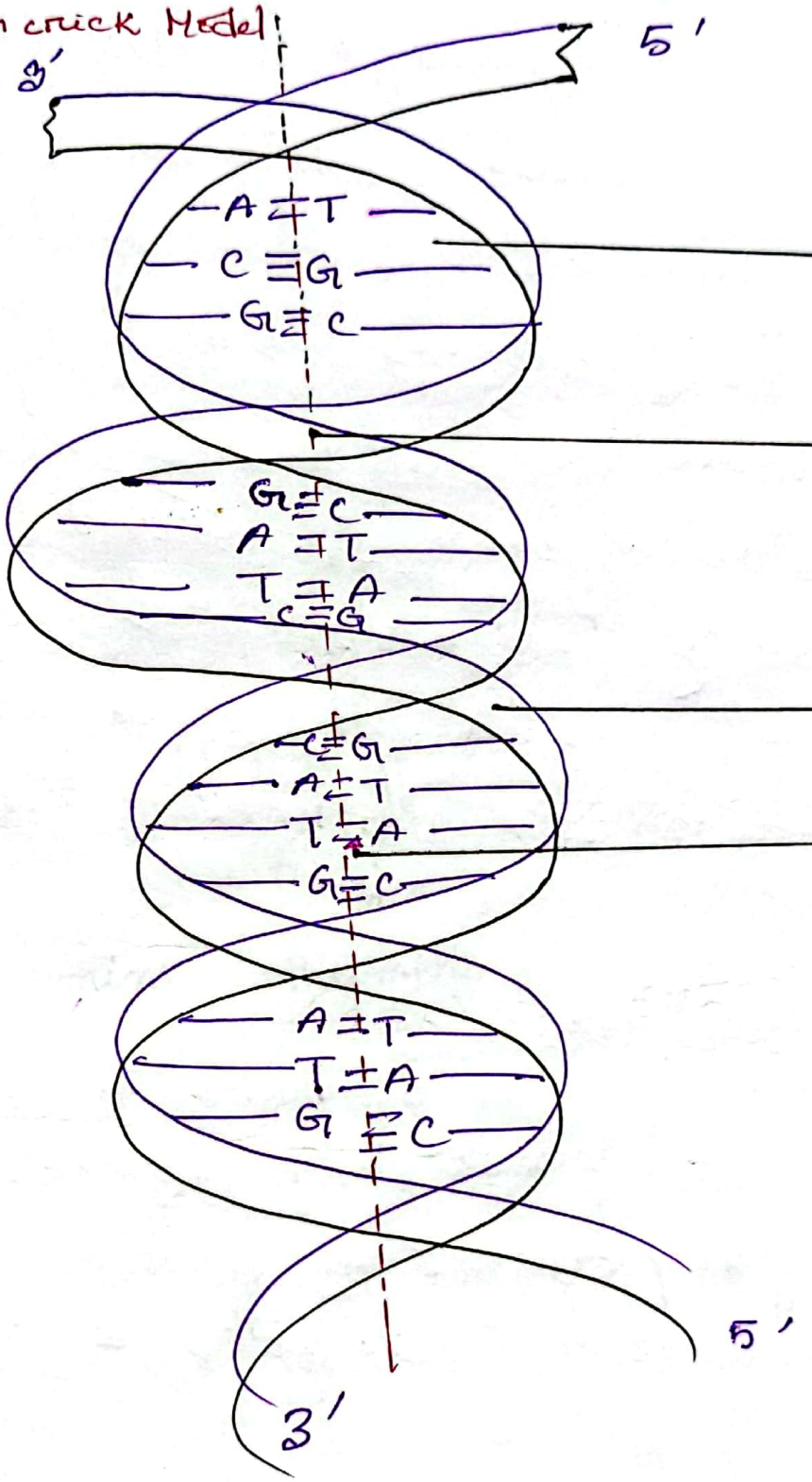
F-20

Double Helix of DNA (Secondary Structure)

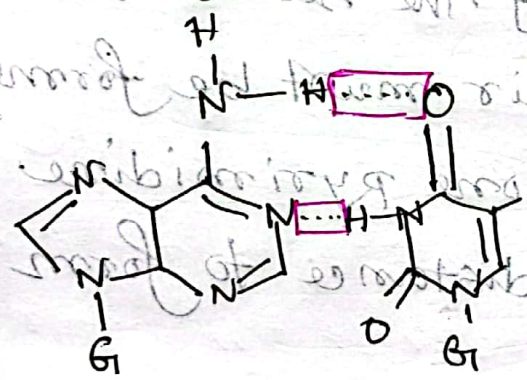
F-21, F-22

Q. Watson crick Model

(4 Marks)



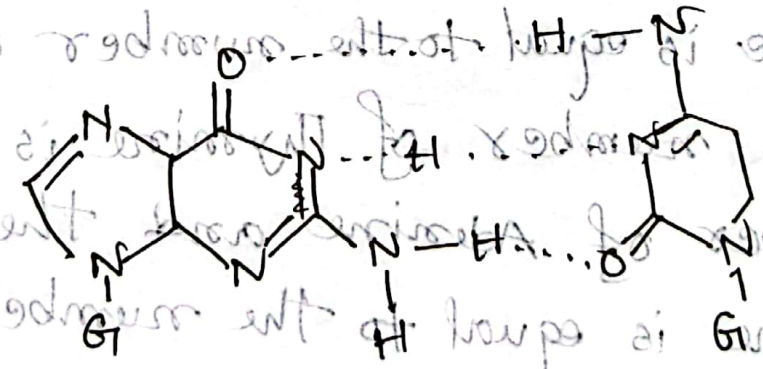
In the double helix, the two chains are coiled around a common axis called the helical axis. The chains are paired in an antiparallel manner i.e the 5' end of one polynucleotide strand is paired with the 3' end of the other polynucleotide strand. One base pair is written in a polynucleotide as (5'-TACG-3')



Two H-bonds.

Adenine Thymine

F-21



Three H-bonds.

Guanine Cytocine

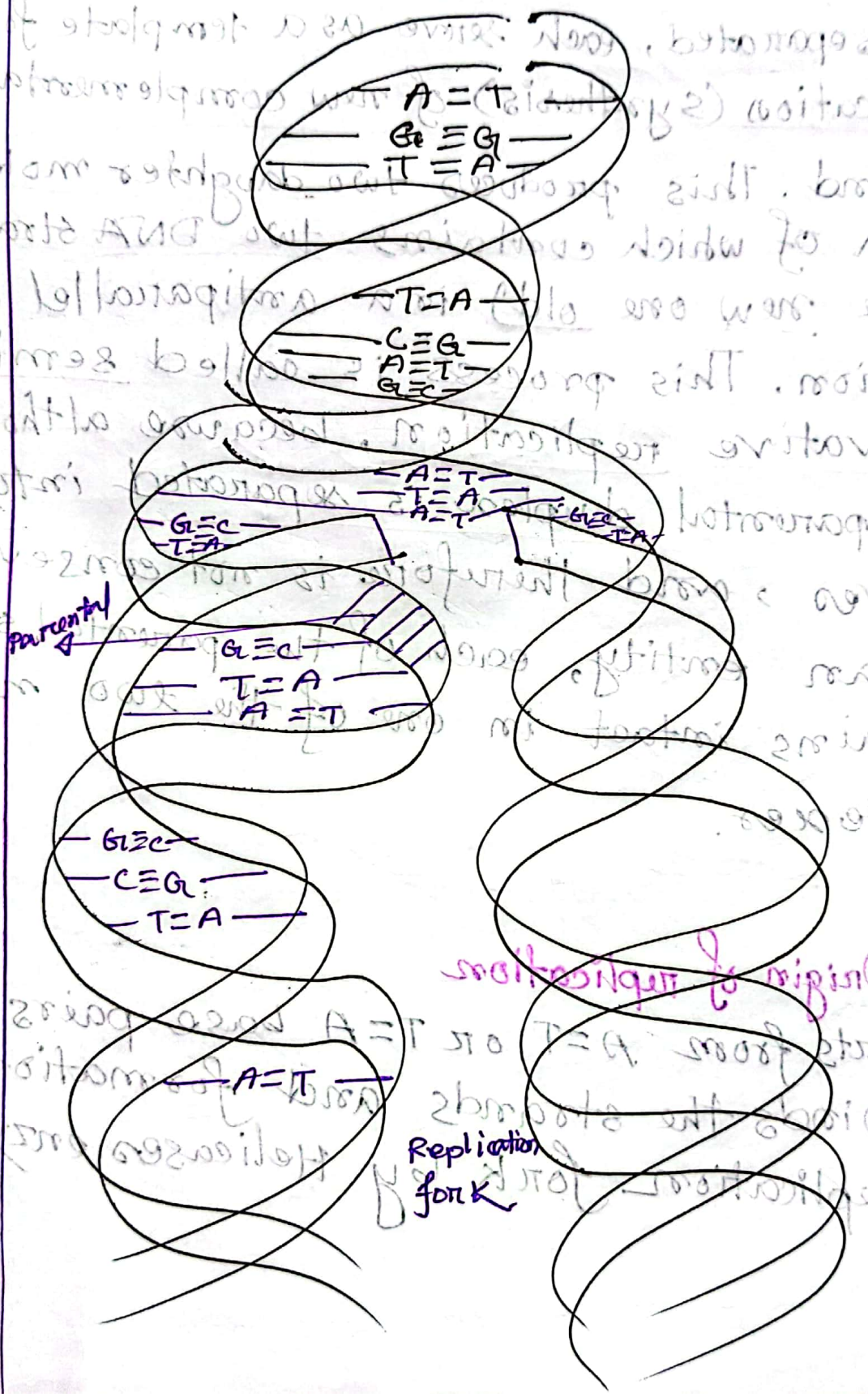
Complementary base pairs in nucleic acid.

In the DNA double helix the hydrophilic deoxyribose-phosphate backbone of each chain is on the outside of the molecule whereas the hydrophobic bases are stacked inside. The overall structure resembles a twisted ladder. The bases of one strand of DNA are paired with the bases of the second strand by H-bonding. The pair must be formed between one purin and one Pyrimidine base to maintain 20 A° distance to form H-bond.

Chargaff rule:

In a nucleotide the number of purin base is equal to the number of pyrimidine base. The number of Thymine is equal to the number of Adenine and the number of Guanine is equal to the number of cytosine. This is called chargaff rule.

Replication of DNA



Replication of DNA:

When the two strands of dsDNA (double stranded DNA) are separated, each serve as a template for the replication (synthesis) of new complementary strand. This produces two daughter molecules, each of which contains two DNA strand (one new one old) in a antiparallel orientation. This process is called semiconservative replication, because although the parental duplex is separated into two halves, and therefore is not conserved as an entity, each of the parental strands remains intact in one of the two new duplexes.

(i) Origin of replication

starts from A=T or T=A base pairs, unwinds the strands and formation of replication fork by Helicases enzyme.

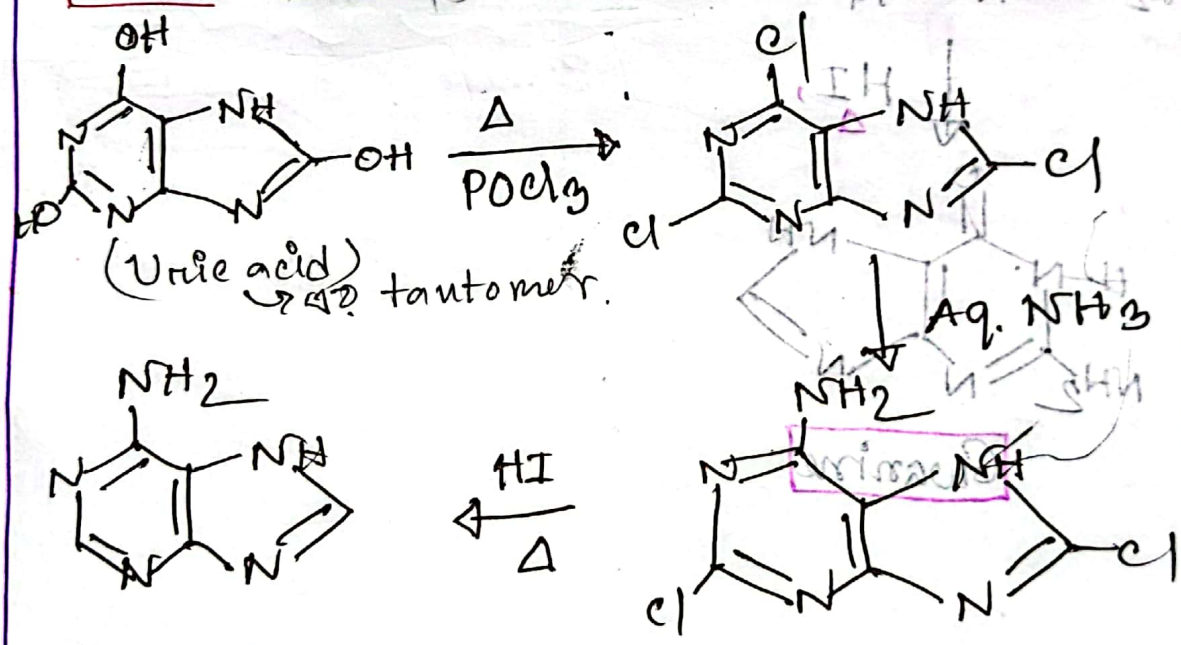
(ii) HDP (Helex destability proteins) don't let the single strand to form double helix.

(iii) DNA direct RNA primer synthesis by RNA polymerase.

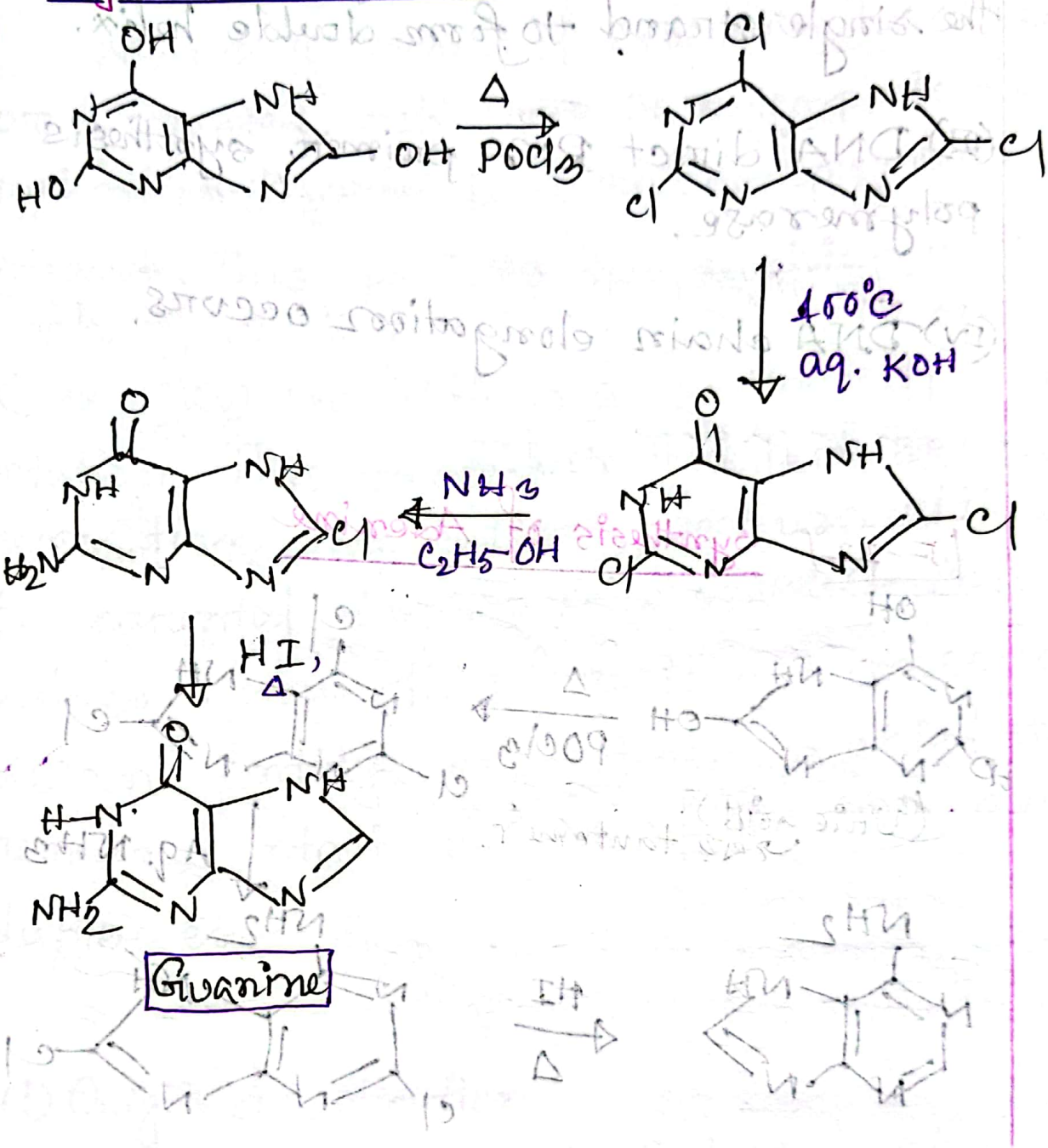
(iv) DNA chain elongation occurs.

F-22

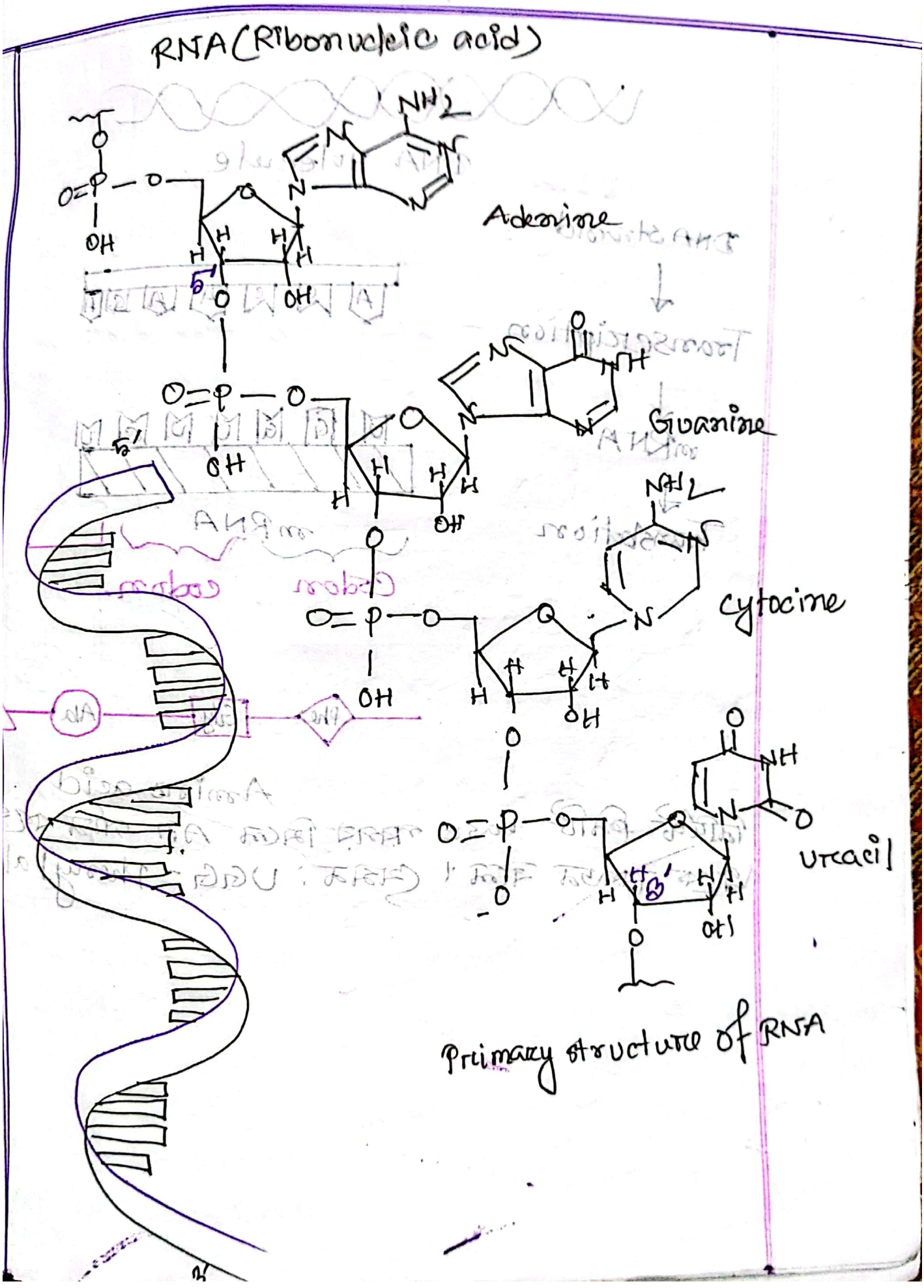
Synthesis of Adenine



Synthesis of Guanine F-22



RNA (Ribonucleic acid)



Primary structure of RNA



DNA Molecule.

DNA strand

↓

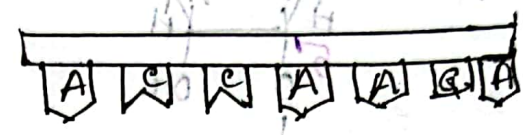
Transcription

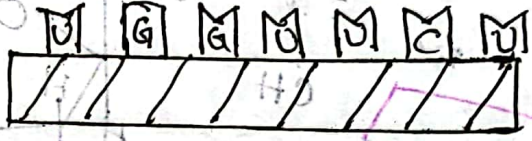
↓

mRNA

↓

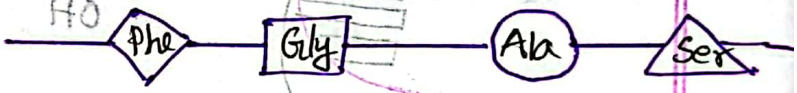
Translation





mRNA

Codon codon.



Amino acid

নির্দিষ্ট তিনটি base পরপর মিলে AA গঠন করে।
একে কোডন বলে। যেমন: UGG = Phenylalanine

Types of RNA

There are three major types of RNA, that participate in the process of protein synthesis, rRNA, tRNA, mRNA. Like DNA these RNA are unbranched polymeric together by 3'—5' phosphodiester bonds.

rRNA (Ribosomal RNA) are found in associated with several protein as components of the ribosomes, the complex structures that serves as the sites for protein synthesis.

tRNA (Transfer RNA) are the smallest of the three major types of RNA molecule. There are at least one specific type of tRNA molecule for each type of the 20 amino acid. 15% of total RNA is tRNA.

mRNA (Messenger RNA)

mRNA comprises only 5% of that RNA in the cell. mRNA is coding RNA in that it carries the genetic information from DNA for the use in protein synthesis.

→F-15

Difference Between RNA vs DNA

DNA	RNA
Sugar moiety is Deoxyribose	Sugar moiety is Ribose
The bases present are Adenine, Guanine, cytocine & Thymine	The bases are Adenine, Guanine, cytocine and uracil.
Double stranded molecule	Single stranded molecules
Obeys chargaff's rule	Does not obey chargaff's rule
Bases are not modified	Bases are modified
It is stable and not hydrolyzed by alkalis	It is unstable and hydrolyzed easily by alkalis.
Life time is comparatively high.	RNA is short lived.
Present in the nucleus, mitochondria and chloroplast	Present in the nucleus, mitochondria, nucleolus, ribosomes and cytosol.

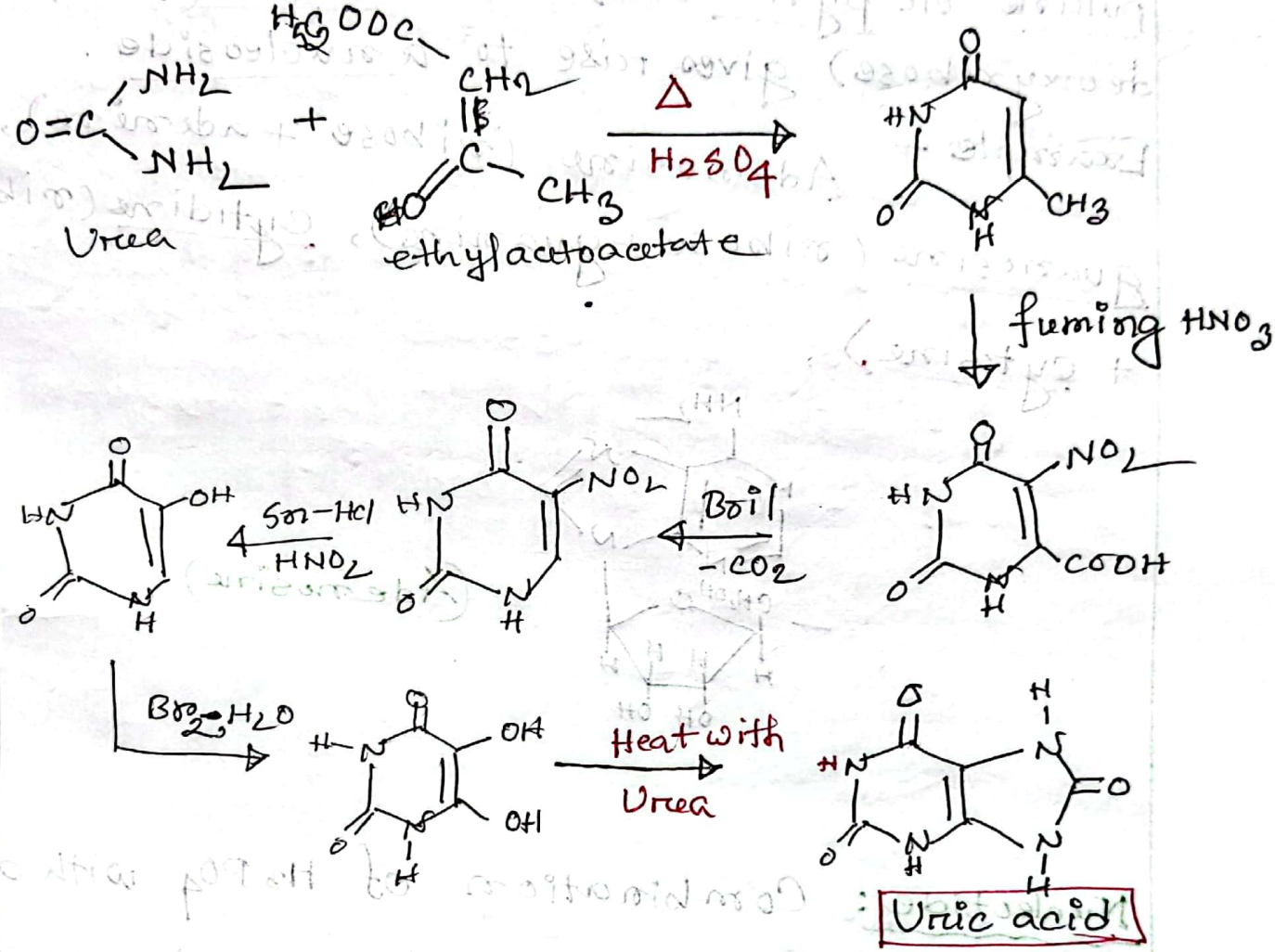
Additional RNA, these RNA performs specialized structural, catalytic and regularly functions and are not translated are called non-coding (RNA).

SUZON

CHEM-50

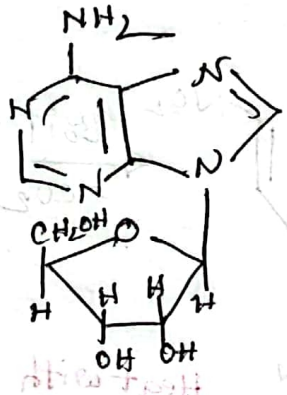
- Why are purines so important? (F-22) (1 Marcle)
- Describe briefly how the structure of nucleotide was established. (4) → F-21
- Define DNA, RNA, Purine, Nucleoside.
- Find out the structure of Uric acid by degradative study.
- Show the relation betn nucleic acid, Nucleosides and nucleotides (2)
- Discuss the importance of purine bases. (2)
- Define Nucleic acid, Nucleotide, Nucleosides.
- Give an out line of DNA replication. (1)
- How can you differentiate ribonucleotide-5'-phosphate from its isomers 2'- and 3'- phosphate? → (1)

Syntheris of Uric acid from Urea and ethylacetoacetate. [F-21]



Nucleosides: Combination of a base (either a purine or pyrimidine) with a sugar (ribose or deoxyribose) gives rise to a nucleoside.

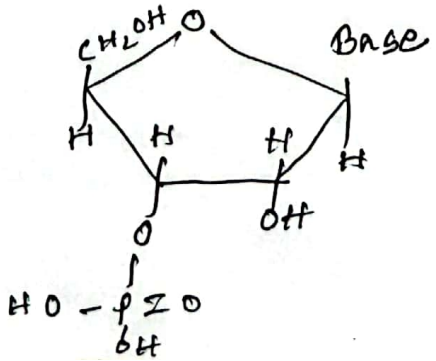
Example – Adenosine (ribose + adenine), guanosine (ribose + guanine), cytidine (ribose + cytosine)



(Adenosine)

Nucleotide: Combination of H_3PO_4 with a nucleoside produces a nucleotide, i.e. nucleotides are nucleoside phosphate.

Ex – Adenylic, guanylic, cytidylic etc.



Ribonucleotide

Nucleic Acid They are colorless, amorphous, high molecular weight polymer, made up of three units

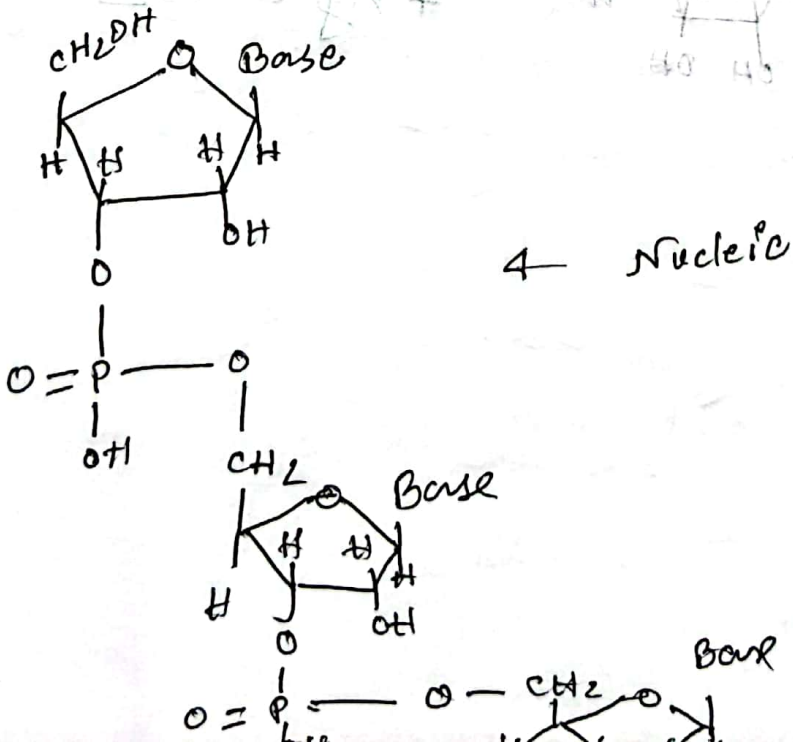
i) A nitrogenous base (Purin or pyrimidine)
ii) Sugars (ribose or deoxyribose)
iii) H_3PO_4 group.

They are obtained by hydrolysis of nucleoprotein.

$$\text{Nucleoprotein} \xrightarrow{H_2O} \text{Nucleic acid} + \text{Protein}.$$

(*) The difference betn nucleotide and nucleic acid is nucleic acids are polymer of nucleotide.

P, B, S

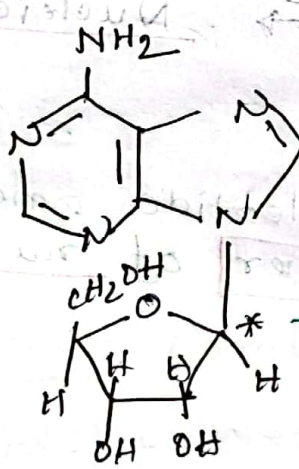


← Nucleic acid

* Why the nucleotide and nucleoside is a non-reducing sugars?

Soln:

In nucleotide and nucleoside, the anomeric -OH group is replaced by the linkage with base, so, it is not free. For this reason, nucleoside and nucleotide does not react with Tollen's reagent to form silver mirror.



→ Anomeric carbon

+ $Ag(NH_3)_2OH$ → No Ag Mirror.

Why are purines so important? (1)
or, Importance of purines (2)

Soln:

Here Key reasons purines are important:

1. Genetic information:
Purines are part of the structure of DNA and RNA, which store and transmit genetic information in cells.

2. Energy transfer:
Purines are involved in energy transfer, notably through ATP, which is primary energy currency.

3. Cell signalling: Adenosine, a derivative of purines, plays a role in various signaling path ways.

4. Metabolism:
Purines are essential for the synthesis of coenzymes like NAD, which are vital for cellular metabolism.

5. Cell growth and Repair:

Purines are involved in the synthesis of nucleotides required for DNA replication and repair, supporting cell growth and tissue regeneration.

Because of these vital roles, any imbalance in purine metabolism can lead to disorders.

Outline of DNA replication: (1 Mark)

1. Initiation
   - (i) Origin of replication
   - (ii) Helicase Activation
   - (iii) single-strand binding protein.
   - (iv) Primase action

2. Elongation
   - (i) DNA polymerase binding
   - (ii) Leading strand
   - (iii) Lagging strand
   - (v) Sliding clamp.

3. Primase Removal and replacement

(i) RNA primer removal

(ii) Ligase action

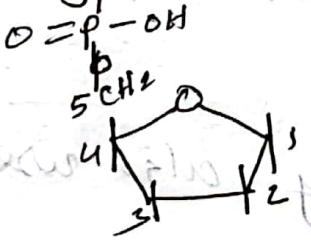
4. Termination:

(i) Replication fork meets

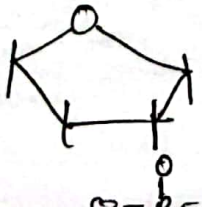
(ii) Proofreading and repair

(*) Differentiate:

*. Ribonucleotide - 5'-phosphate:
The phosphate group is attached to the 5' carbon of the ribose sugar.

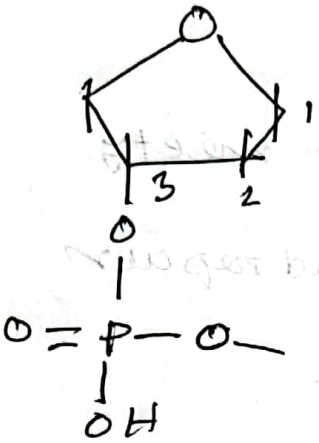


* Ribonucleotide - 2'-phosphate: The phosphate group is attached to the 2' carbon of the ribose sugar.



* Ribonucleotide - 3'- phosphate:

The phosphate group is attached to the 3' carbon of the ribose sugars.



Method:

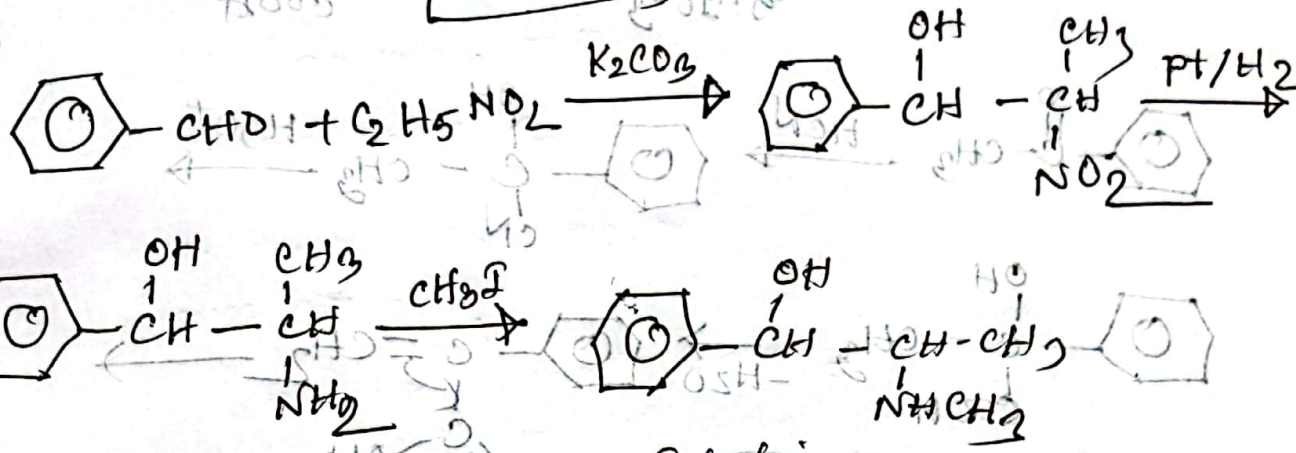
NMR spectroscopy can directly reveal the positions of the phosphate groups based on the chemical shifts and correlation betn phosphate and sugars carbons.

Mass spectrometry also used (m/z ratios)

Synthesis of Ephedrine

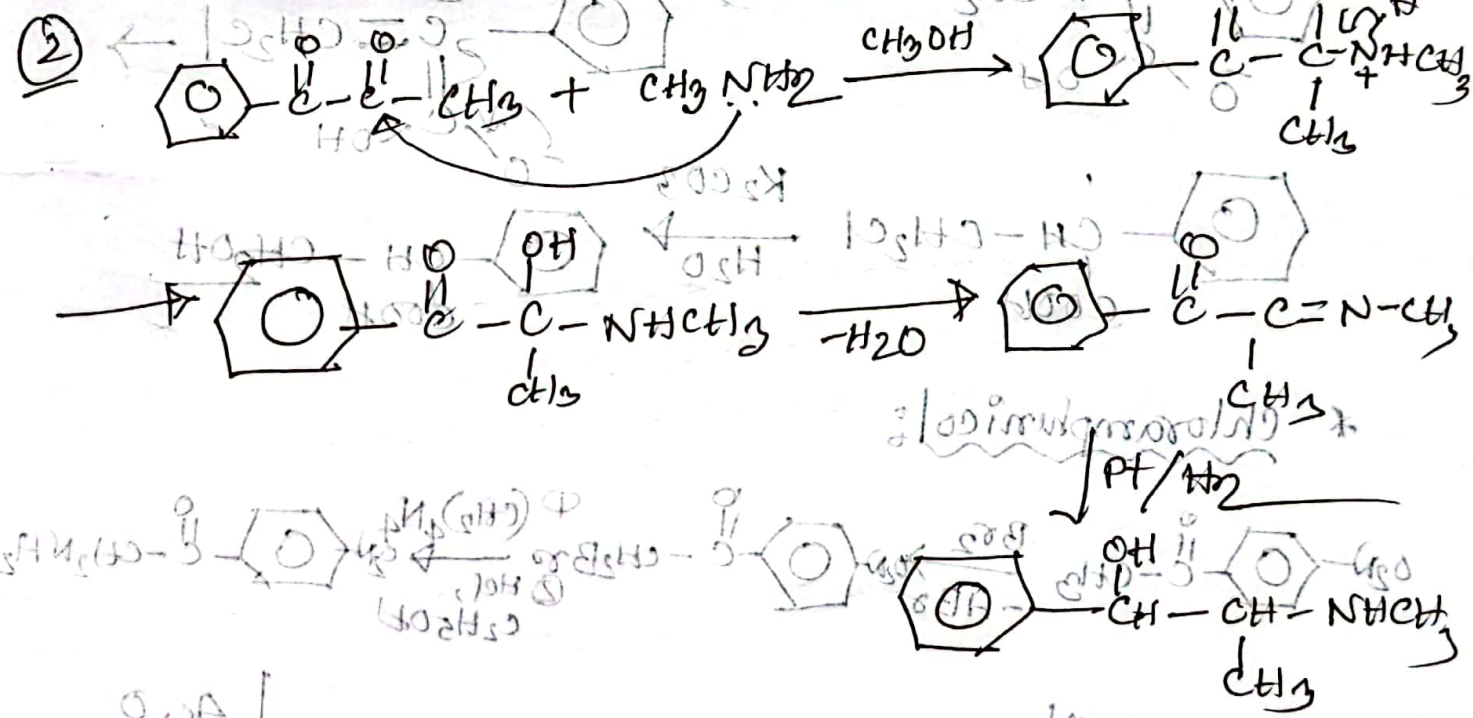
$C_9H_{15}NO$

1.



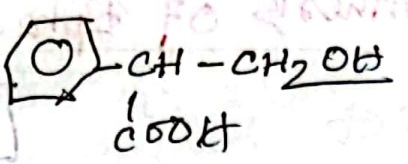
Ephedrine.

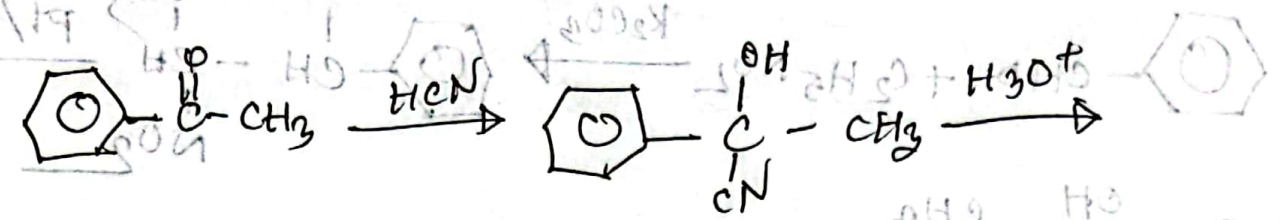
2.

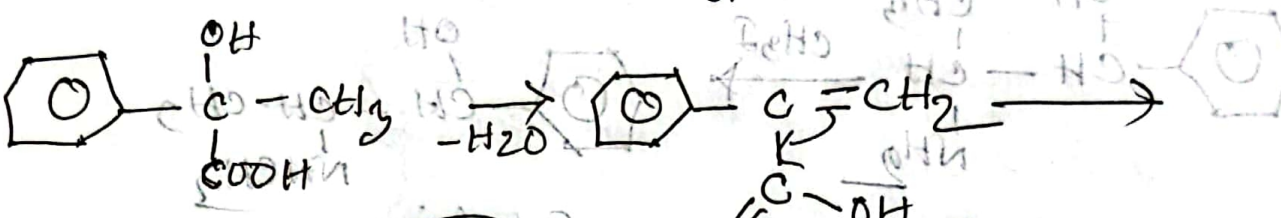


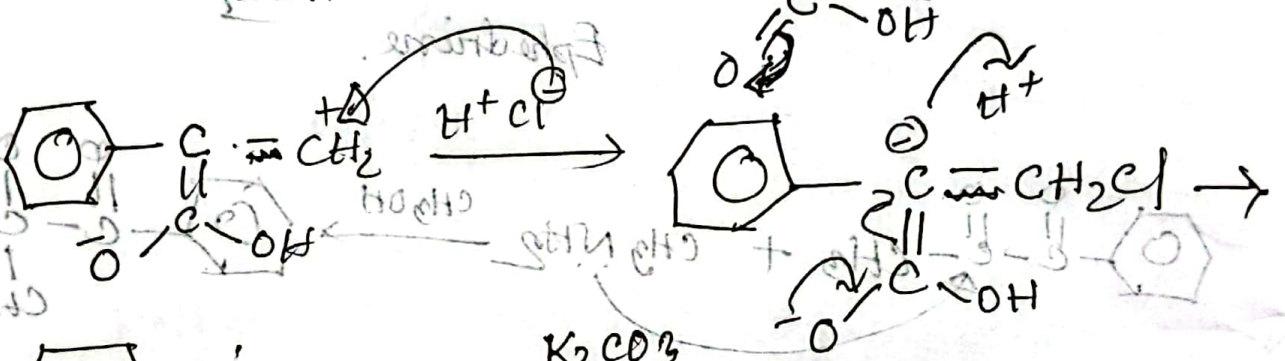
(*) Synthesis of tropic Acid:

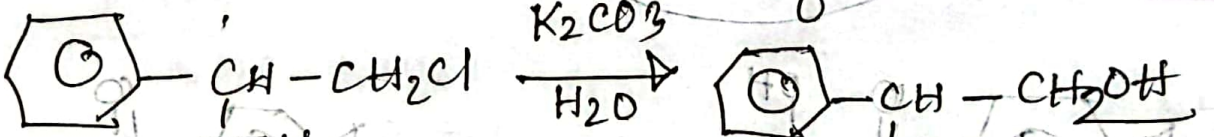
$C_9H_{10}O_3$



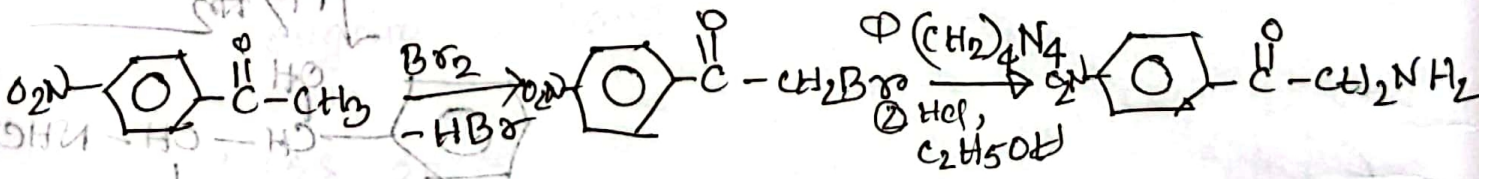


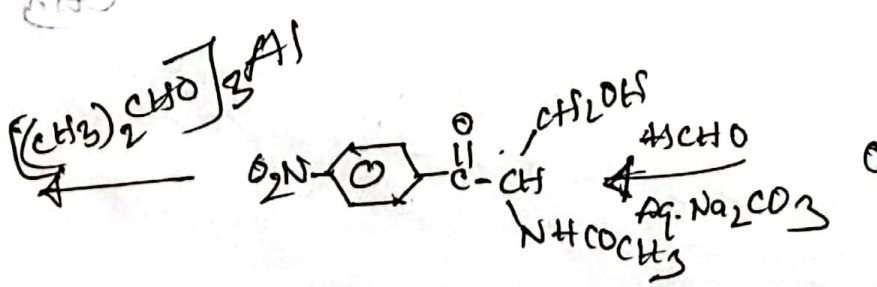


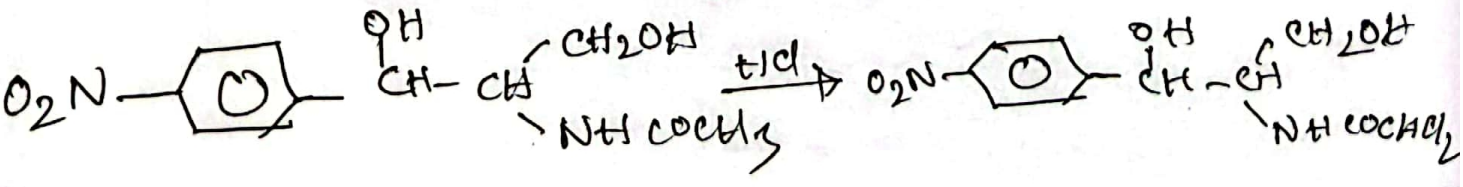




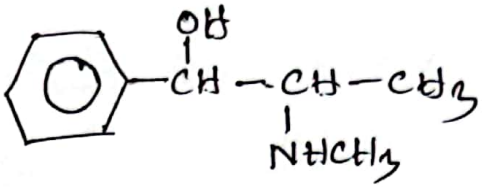
* Chloramphenicol:



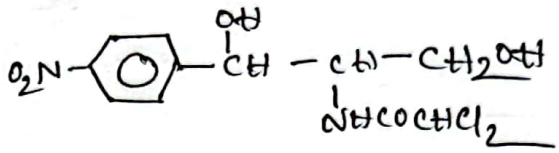




Ephedrine : $C_{10}H_{15}NO$



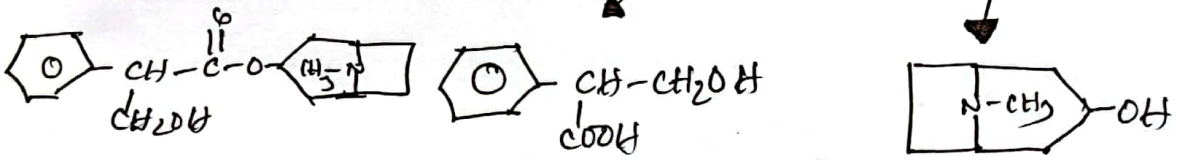
Chloramphenicol : $C_{11}H_{12}Cl_2N_2O_5$



Atropine :

$C_{17}H_{23}NO_3 \rightarrow$ Tropic acid + Tropine

$C_9H_{10}O_3$ + $C_8H_{15}NO$



Penicillin : $C_9H_{11}N_2O_4SR \rightarrow C_5H_{11}NO_2S + C_8H_4NO_2R + CO_2$

Penicillamine